The Rat Race at Work: Technology and Burnout

Auyon Siddiq
UCLA Anderson School of Management

October 2026

Abstract

Artificial intelligence can raise employee productivity, but empirical evidence suggests it may also intensify work. This paper studies how productivity-enhancing technology impacts employee welfare in competitive workplaces, accounting for employees' strategic work decisions and risk of burnout. We develop a model in which employees' productivity increases in both technology quality and their chosen work intensity. Higher work intensity raises the hazard rate of burnout, at which point the employee's output falls to zero. We show that when employees' rewards depend sufficiently strongly on their output relative to peers, better technology can lead to a competitive "rat race" that induces more work. As a result, employee welfare can initially decrease in technology quality due to intensified competition, but eventually rebound as competition cools and employees reap the productivity benefits of the technology. For the firm, the possibility of burnout can lead to decreasing marginal returns to technology; in the absence of burnout, firm profit is convex in technology quality. When the firm chooses employee performance incentives, a reduction in the underlying burnout hazard can lead to a higher risk of burnout and lower employee welfare in equilibrium. Finally, when technology quality is low, a profit-maximizing firm may invest more in burnout mitigation than a social welfare-maximizing firm, which instead promotes welfare by suppressing workplace competition through weaker performance incentives.

1. Introduction

Technology is transforming the workplace: AI assistance has been shown to increase the quality or volume of employee output across several types of knowledge work, including professional writing, management consulting, customer support, and software development (Noy and Zhang, 2023; Dell'Acqua et al., 2023; Brynjolfsson et al., 2025; Cui et al., 2025).

Improvements in employee productivity, however, need not translate into lighter workloads. In fact, recent empirical evidence suggests AI may be having the opposite effect: Jiang et al. (2025) find that workers in occupations with greater AI exposure work longer hours, with stronger effects when AI complements employee work. Similarly, Ranganathan and Ye (2026) find that employees at a technology firm took on more tasks and worked longer hours following the adoption of AI – often without being asked – and a study of workplace activity data finds that time spent on email and

messaging increased sharply after AI adoption (ActivTrak Productivity Lab, 2026).

Why might a technology that boosts productivity lead to *more* work? One explanation is competition: When employees are evaluated based on their performance relative to peers, better technology can make production easier for everyone, leading employees to work harder for the same rewards.[1] This type of strategic overwork can create a "rat race" in which employees work harder to preserve their position relative to coworkers, even though they would collectively prefer a less demanding equilibrium (Nalebuff and Stiglitz, 1983; Landers et al., 1996). OpenAI CEO Sam Altman makes a similar prediction about how AI will affect work: "It's like a relative game. People are very focused on how they're doing relative to other people . . . I think we're all going to be much busier than we thought we were supposed to be" (Altman, 2026).

This paper asks two questions. First, how does productivity-enhancing technology affect employee welfare in competitive workplaces? Second, to what extent can improved work conditions protect employees from the costs of this competition?

Evaluating employees relative to their peers is common, particularly in the technology sector. In 2025, Meta set a target for teams with at least 150 employees to place 15 to 20 percent of employees in the lowest performance category, and described the review process as an opportunity to make exit decisions (Dixit and Mann, 2025). Microsoft and Amazon have also used similar ranking systems (Eichenwald, 2012; Soper, 2022). However, ranking-based systems are not necessary for employees to compete: Bonuses, raises, and promotions are often limited, so employees may recognize that compensation and career advancement implicitly depend on how their performance compares with coworkers'. In such settings, an across-the-board productivity gain from technology can make workplace competition more intense, even when the employer leaves its performance evaluation system unchanged.

Whether better technology compels employees to work more is consequential because higher workloads impose costs on both employers and employees. In particular, excessive work may lead to employee *burnout*, which refers to the exhaustion and reduced effectiveness caused by chronic workplace stress (World Health Organization, 2019; Maslach et al., 2001). For the firm, burnout can reduce productive capacity through increased sickness absence or employee departures (Corbeanu et al., 2023; Schaufeli et al., 2009; Salvagioni et al., 2017). One estimate places the annual cost of burnout at approximately $5 million for an average U.S. company with 1,000 employees, largely through reduced productivity (Martinez et al., 2025). Concerns that AI adoption may exacerbate burnout are growing in both the popular press and academic research, which have proposed a variety of mechanisms (Upwork Research Institute, 2024; Torrence and Mascarenhas, 2026; Liu et al., 2024). Technological progress

[1] Nalebuff and Stiglitz (1983) note the possibility that productivity-enhancing technology might intensify workplace competition, although they do not formally study it.

can therefore create an operational tradeoff: It can raise individual employee productivity, but it also risks increasing burnout, which erodes the firm's productive capacity.

In recognition of the costs imposed by burnout, many employers explicitly invest in employee well-being, with workplace wellness programs now covering more than 50 million U.S. workers (Jones et al., 2019). For example, HubSpot combines coaching, therapy, and meditation in what it calls a "long-term framework to prevent and battle burnout" (HubSpot, 2021), and Salesforce and Adobe reimburse employees for activities that support mental well-being (Hyder, 2022; Adobe, 2026). Many companies view employee well-being as a strategic priority: In a Deloitte survey, 83 percent of executives said well-being should be discussed by the board (Fisher et al., 2023). Beyond wellness programs, employers can also reduce burnout by changing the work environment itself, for example, by giving employees greater flexibility over their work schedules (Moen et al., 2016; Kelly and Moen, 2020). Changes to organizational culture, such as improved support from supervisors and coworkers, may also weaken the link between job demands and burnout (Bakker et al., 2005). Whether through wellness programs or changes to the work environment, burnout mitigation may become increasingly important if technological improvements indeed intensify work.

1.1. Summary of Model and Results

We develop a single-period, continuous-time model of a firm with one team of n identical employees. All employees simultaneously choose their work intensities at the start of the period. Employee output increases in their work intensity, duration of work, and technology quality. Employees face relative-performance incentives, which reward them based on both their absolute output and output relative to the team average. We model employee burnout through a stochastic time-to-failure process in which greater work intensity raises the hazard rate of burnout. A burnout event imposes a fixed cost on the employee and ends their production for the remainder of the period. The firm's payoff is total output less employee payments, making burnout costly for the firm through reduced production. We consider three settings: one in which the strength of the relative-performance incentive is fixed, one in which the firm chooses it, and one in which the firm jointly chooses the incentive and invests in burnout mitigation.

We first examine how technology *quality* – the magnitude of the boost it provides to employee output – affects employee welfare when the relative-performance incentive is fixed. When technology quality is modest, a further improvement creates a "rat race effect" that erodes employee welfare: Employees work more intensely to keep up with coworkers who they anticipate will become more productive, and the resulting strain on employees outweighs the productivity gains, leaving them collectively worse off. However, when quality is high, employees may benefit from further improvements. Intuitively,

better technology amplifies the performance advantage of the employees who remain productive when a coworker burns out, allowing them to earn large relative-performance rewards simply by avoiding burnout. This "strategic survival" behavior weakens the marginal incentive to work intensely, leading equilibrium work intensity to rise more slowly – or decline – in technology quality. This cooling of workplace competition allows employees to enjoy the direct productivity benefit of better technology and employee welfare to rebound.

We next consider how the firm's returns to technology are shaped by employees' strategic work decisions. Without burnout, better technology both raises the output per unit of employee work and induces employees to work more intensely. Both of these effects combine to make firm profit convex in technology quality. With the possibility of burnout, however, firm returns to technology may instead be decreasing when relative-performance incentives are strong, due to the strategic moderation of work intensity described above. This result lends additional theoretical support to the idea that the organizational environment of a firm can shape how it benefits from technology (Brynjolfsson and Hitt, 2000), in our case through the effects of employee incentives and well-being.

If technology intensifies work, then employees might look to improved work conditions (e.g., flexible hours) for protection from burnout. However, we show that when performance incentives are chosen by the employer, a reduction in the underlying burnout hazard (which we call burnout *exposure*) can lead to a higher risk of burnout and lower employee welfare in equilibrium. The intuition is that a lower hazard keeps employees productive for longer at a given work intensity, making the employer's productive capacity more secure. The employer's response is then to demand even more work from employees, which it does by strengthening the relative-performance incentive, thereby intensifying competition. When burnout exposure is sufficiently small, the resulting increase in employee work intensity more than offsets the direct reduction in the hazard, making employees more likely to burn out. The stronger performance incentive does not offset these costs, and employee welfare falls despite the improvement in work conditions. Improvements in work conditions can thus leave employees worse off when the employer can freely adjust incentives.

Finally, we consider a setting where the firm jointly chooses the employees' relative-performance incentive and an investment in burnout mitigation (e.g., wellness programs) that lowers the underlying burnout hazard rate. We benchmark the optimal design of a profit-maximizing firm against that of a welfare-maximizing firm, which maximizes the sum of firm profit and employee welfare. When technology quality is low, the profit-maximizing firm invests *more* in burnout mitigation than the welfare-maximizing firm, which instead limits burnout by suppressing employee competition through a weaker performance incentive. Because moderating competition sacrifices output, this result holds at low technology levels where the output loss is limited; when technology quality is high, we find

numerically that the welfare-maximizing firm invests more in mitigation than the profit-maximizing firm. These findings suggest that mitigation spending alone can be a misleading indicator of how well a workplace protects its employees. Further, when the gains from technology are modest, it may be more welfare-efficient for the firm to weaken relative-performance incentives than to spend aggressively on mitigation.

1.2. Related Literature

This paper relates to the people-centric operations literature, which studies the links between employee behavior, well-being, and operational performance (Boudreau et al., 2003; Roels and Staats, 2021; Corbett, 2024). This line of work has shown that workers adjust their pace to system load and coworkers' pace (Schultz et al., 1998; KC and Terwiesch, 2009), and that excessive workloads can reduce productivity and service quality (Tan and Netessine, 2014; Kuntz et al., 2015; Berry Jaeker and Tucker, 2017) or increase absenteeism (Green et al., 2013). Prior work has also examined how fatigue, supplier audits, and process-safety improvements affect worker performance and well-being (Bavafa and Jónasson, 2024; Jiao et al., 2026; Bakshi and Peura, 2022). We extend this literature by focusing on how technological progress can impact employee welfare in the workplace.

We also connect to a rich literature on the use of relative performance evaluation to motivate employee effort, which includes rank-order tournaments (Lazear and Rosen, 1981; Green and Stokey, 1983; Connelly et al., 2014) and more general incentive contracts (Nalebuff and Stiglitz, 1983; Mookherjee, 1984). Green and Rahmani (2022) compare how relative and absolute rating systems shape employee effort and worker payoffs. Performance-based pay has been found to raise worker output (Lazear, 2000), and feedback on relative performance can change how much employees work (Casas-Arce and Martínez-Jerez, 2009; Song et al., 2018). This paper is closest to studies of the unintended consequences of performance incentives, which can encourage workers to undermine coworkers (Lazear, 1989), increase stress and burnout (Habel et al., 2021), or push workers to overexert (Landers et al., 1996; Greiner et al., 2026).

Our model is motivated by prior empirical work on employee burnout, which has identified workload and job demands as the main factors (Cordes and Dougherty, 1993; Maslach et al., 2001; Schaufeli and Bakker, 2004), and shown that improvements in employee health may boost productivity (Gubler et al., 2018). Burnout is costly for both employees and firms and is associated with worse job performance (Corbeanu et al., 2023), and workplace stress is associated with significant healthcare costs (Goh et al., 2016). Improved work conditions, such as greater employee autonomy and strong supervisor relationships, can buffer the effect of job demands on burnout (Bakker et al., 2005). Our model captures this distinction by separating employees' chosen work intensity from the underlying burnout

hazard, which reflects work conditions and past investments in burnout mitigation. Prior analytical models have examined how requests from managers can lead to burnout (Matros et al., 2026), and how overwork can drive attrition (Waiser et al., 2025).

More broadly, a growing empirical literature examines how new workplace technologies affect employee welfare. AI assistance has been shown to improve task satisfaction and lower employee attrition (Noy and Zhang, 2023; Brynjolfsson et al., 2025). However, new technologies can also raise work demands, leading to more overtime and greater work intensity (Giebel and Lammers, 2025), expectations of constant availability (Mazmanian et al., 2013), and longer workdays (Jiang et al., 2025).

Lastly, this paper relates to the operations management literature on how AI changes the nature of work. This literature has examined how the benefits of AI depend on how tasks are allocated between humans and AI, how their interaction is designed, and their relative capabilities (Fügener et al., 2026; Fang et al., 2026; Hu et al., 2026; Feldman et al., 2026). Dizdarer et al. (2025) study how AI changes firms' mix of employees and contractors. AI assistance can increase the cognitive effort and time pressure that workers face (Boyacı et al., 2024, 2026; Legros and de Véricourt, 2026). Other work studies how AI interacts with worker incentives: Dai and Taylor (2026) examine how firms jointly design AI systems and employee incentives, and Lu and Tomlin (2026) and Guan et al. (2025) show that limiting AI disclosure or precision can preserve workers' incentives to acquire information. Bastani and Cachon (2025) show that more reliable AI can make human oversight more costly to induce, and Siderius et al. (2026) show that better AI can weaken workers' incentives to check its output, causing skill atrophy. Much of this work has focused on a single employee; we extend this by abstracting away from the details of how AI drives task-level productivity and focusing on how it changes strategic behavior among multiple employees.

2. Model

Consider a firm with a single team of $n \geq 3$ ex ante identical employees, indexed by $i = 1, \ldots, n$. Team size n is fixed. There is one period, normalized to unit length, with continuous time $t \in [0, 1]$. At the beginning of the period, all employees simultaneously choose their work intensities $x_i \in [0, \infty)$, where x_i measures work effort supplied per unit of time while productive. Higher intensity may reflect longer working hours, greater effort per hour, or both.

Let $\tau_i(x_i) \in (0, 1]$ denote the time for which employee i remains productive during the period. If employee i does not burn out during the period, then $\tau_i(x_i) = 1$. If $\tau_i(x_i) < 1$, employee i stops producing for the rest of the period, with no recovery or reassignment of work. This captures episodes such as extended sickness absence or disengagement (Schaufeli et al., 2009; Cordes and Dougherty,

1993).[2] Completed work is therefore $x_i\tau_i(x_i)$.

2.1. Technology Quality and Performance Incentives

All employees use a common technology of quality $\theta \in [1, \theta^m]$, where $\theta^m > 1$ is an upper bound on quality. We assume throughout that θ is determined by external forces (e.g., market-wide advances in AI), and is thus exogenous. Technology multiplies the output produced by work: Employee i's realized output as a function of work intensity is $y_i(x_i) \equiv \theta x_i \tau_i(x_i)$, and $\mathbf{y}(\mathbf{x}) = (y_1(x_1), \ldots, y_n(x_n))$ is the realized output profile across workers. This multiplicative specification follows Nalebuff and Stiglitz (1983) and is also consistent with an O-ring production technology (Kremer, 1993), which has been widely used to model productivity gains from AI (Gans and Goldfarb, 2025).

The employee is paid for their realized output y_i; we normalize payment per unit of output to 1. The firm values each unit of output at $v > 1$. We also allow compensation to depend on performance relative to coworkers, as in the tournament and multi-agent contracting literature (Lazear and Rosen, 1981; Nalebuff and Stiglitz, 1983; Mookherjee, 1984). We do so by including a relative-performance component of employee pay, r_i, so the firm's total payment to the employee is $y_i(x_i) + r_i$.

To define r_i, let $\bar{y} = n^{-1} \sum_j y_j$ denote average output across the team, and let $\gamma \geq 0$ denote the strength of the relative-performance incentive; equivalently, the output advantage at which relative-performance pay reaches its maximum. Employee i's relative-performance pay is then

$$r_i(\mathbf{y}) = \int_0^{y_i - \bar{y}} (\gamma - z)_+ \, dz, \tag{1}$$

where $(z)_+ \equiv \max\{z, 0\}$. We use (1) as a tractable (i.e., concave) approximation to relative pay schedules used by firms. The function r_i rewards performance relative to the team average, capped at a level that depends on γ. A larger γ thus rewards high performers more aggressively. For simplicity, we refer to γ as the *incentive*. All parties are risk neutral.

2.2. Work and Burnout

Employee i incurs instantaneous work cost $x_i^\alpha t^{\alpha-1}$ while productive, where $\alpha > 2$.[3] This cost function captures the increasing disutility of work: Maintaining a given pace becomes more burdensome for the employee as work continues. We assume the burnout hazard is proportional to instantaneous work cost: An employee working at intensity x_i burns out at hazard rate $\rho x_i^\alpha t^{\alpha-1}$, with $\rho > 0$. We call

[2] We conjecture that milder forms of burnout, in which an employee continues to produce at reduced quality, would temper the main results but not change them qualitatively.

[3] The restriction $\alpha > 2$ is a technical condition that ensures that the firm's optimal induced work intensity in Sections 4 and 5 is finite.

ρ the burnout *exposure* parameter, which represents the prevailing work conditions and scales the hazard rate. Exposure is common across employees to reflect shared work conditions within the team.

This specification of the burnout hazard leads to a Weibull time-to-failure model (Weibull, 1951), commonly used in reliability engineering to model mechanical failure. We censor burnout time at the end of the period: for $t \in [0, 1]$,

$$\Pr(\tau_i(x_i) < t \mid x_i) = 1 - \exp\left\{-\int_0^t \rho x_i^\alpha s^{\alpha-1}\, ds\right\} = 1 - e^{-\rho(x_i t)^\alpha/\alpha}.$$

Employees who remain productive throughout the period have $\tau_i(x_i) = 1$. The mean burnout time on the team is thus $\mathbb{E}[\tau_i(x_i)]$. For fixed α, a lower mean burnout time corresponds to a higher probability of burnout during the period. The variables $\tau_i(x_i)$ are independent across employees conditional on work intensity x_i. Figure 1 shows the cumulative burnout probability for different levels of exposure and work intensity, holding the shape parameter $\alpha = 3$ fixed.

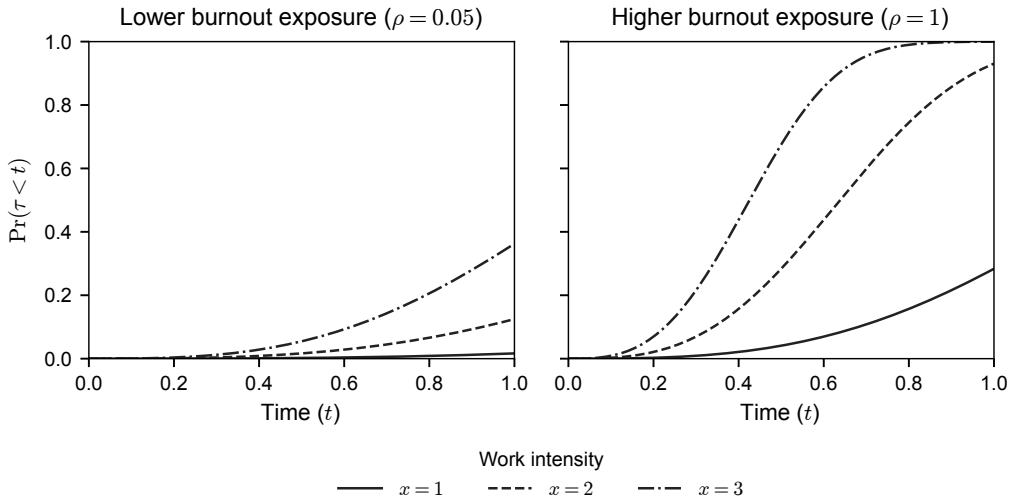


Figure 1: Cumulative distribution functions of random burnout time under varying burnout exposure levels (ρ) and work intensities (x), with $\alpha = 3$.

Burnout imposes an employee cost $\phi^w > 0$, which captures its health and career consequences (Goh et al., 2016; Salvagioni et al., 2017). A burnout episode also imposes a burnout cost $\phi^f > 0$ on the firm, which captures expenses such as employer-funded counseling or administrative costs associated with an employee's absence or departure. All model parameters are common knowledge.

2.3. Equilibrium

An employee's total expected work cost over the period is $\mathbb{E}\left[\int_0^{\tau_i(x_i)} x_i^\alpha t^{\alpha-1}\, dt \mid x_i\right]$. Taking coworkers' intensities $\mathbf{x}_{-i}$ as given, employee i chooses x_i to maximize expected compensation net of work and

burnout costs:

$$u_i(x_i, \mathbf{x}_{-i}) \equiv \mathbb{E}\Bigg[\underbrace{y_i(x_i) + r_i(\mathbf{y}(\mathbf{x}))}_{\text{Absolute and relative pay}} - \underbrace{\int_0^{\tau_i(x_i)} x_i^{\alpha} t^{\alpha-1}\, dt}_{\text{Work cost}} - \underbrace{\phi^w \mathbf{1}\{\tau_i(x_i) < 1\}}_{\text{Burnout cost}} \;\Bigg|\; \mathbf{x} \Bigg]. \tag{2}$$

We generally suppress dependence on θ, ρ, and γ. Here, $\mathbf{1}\{\tau_i(x_i) < 1\}$ equals one if employee i burns out and zero otherwise. A work intensity that maximizes this expected payoff is a best response to $\mathbf{x}_{-i}$. Total expected employee welfare for an arbitrary intensity profile is then $\sum_{i=1}^n u_i(x_i, \mathbf{x}_{-i})$. The firm's profit is output value less employee payments and burnout costs. Its expected value is

$$\pi(\mathbf{x}) \equiv \mathbb{E}\Bigg[\sum_{i=1}^{n} \Big(\underbrace{v y_i(x_i)}_{\text{Output value}} - \underbrace{\big[y_i(x_i) + r_i(\mathbf{y}(\mathbf{x}))\big]}_{\text{Absolute and relative pay}} - \underbrace{\phi^f \mathbf{1}\{\tau_i(x_i) < 1\}}_{\text{Burnout cost}} \Big) \;\Bigg|\; \mathbf{x} \Bigg]. \tag{3}$$

A pure-strategy Nash equilibrium is an intensity profile in which every employee's intensity is a best response to coworkers' intensities. Formally, $\mathbf{x}^*$ is an equilibrium if $u_i(x_i^*, \mathbf{x}_{-i}^*) \geq u_i(z, \mathbf{x}_{-i}^*)$ for all $z \geq 0$ and $i = 1, \ldots, n$. Intensities are chosen before burnout is realized, so outputs and payoffs remain random at the time of choice.

Lemma 1. *The employees' work-intensity game has a unique and symmetric Nash equilibrium. The equilibrium intensity x^* is the unique solution to*

$$(1 + \rho\phi^w) x^{\alpha-1} = \theta + \frac{n-1}{n} \theta\, \mathbb{E}\left[\left(\gamma - \frac{\theta x}{n} \sum_{j=1}^{n-1} [1 - \tau_j(x)] \right)_+ \right]. \tag{4}$$

The equilibrium condition balances the pay from additional work against its work and burnout costs. The term $x[1 - \tau_j(x)]$ captures the work lost when coworker j burns out, which gives employees who remain productive a relative-performance advantage. We write the symmetric equilibrium profile as $\mathbf{x}^* = (x^*, \ldots, x^*)$, and the equilibrium mean burnout time as $\mathbb{E}[\tau_i(x^*)]$. At this equilibrium profile, summing (2) gives equilibrium employee welfare $W^* \equiv \sum_{i=1}^n u_i(x_i^*, \mathbf{x}_{-i}^*)$, while (3) gives firm profit $\Pi^* \equiv \pi(\mathbf{x}^*)$.

3. Technology, Employee Welfare, and Firm Profit

We begin by considering how improvements in technology quality θ influence employees' work intensity (Section 3.1), employee welfare (Section 3.2), and firm profit (Section 3.3). We hold the incentive γ fixed in this section.

3.1. Impact of Technology on Work Intensity

How employees adjust their work intensity is central to how technology affects both employees and the firm. Lemma 2 characterizes how employees' work-intensity response to improving technology depends on the incentives they face.

Lemma 2 (Rat race effect)**.** *Let $\bar{\gamma}_R \equiv n/(n-1)$. The following statements hold.*

(i) If $\gamma \in [0, \bar{\gamma}_R]$, then equilibrium work intensity x^* strictly increases and mean burnout time $\mathbb{E}[\tau_i(x^*)]$ strictly decreases in technology quality θ on $[1, \theta^m]$.

(ii) Suppose $\theta^m > 1 + \bar{\gamma}_R$. For each $\gamma \in (\bar{\gamma}_R, \theta^m - 1)$, for sufficiently large ρ, there are burnout-cost thresholds $0 \leq \underline{\phi}^w < \overline{\phi}^w$ with the following property: If $\phi^w \in (\underline{\phi}^w, \overline{\phi}^w)$, then x^* strictly decreases and $\mathbb{E}[\tau_i(x^*)]$ strictly increases in θ on $[\bar{\theta}, \theta^m]$ for some $\bar{\theta} \in (1, \theta^m)$.

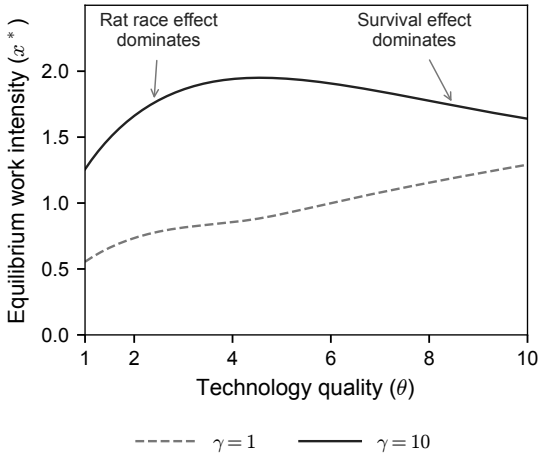


Figure 2: Equilibrium work intensity for varying technology qualities under weaker ($\gamma = 1$) and stronger ($\gamma = 10$) relative-performance incentives. Dashed and solid lines depict behavior described in parts (i) and (ii) of Lemma 2, respectively. Parameters: $\alpha = 3$, $n = 10$, $\rho = 10$, $\phi^w = 0.5$, $\theta^m = 12$.

Figure 2 illustrates the two cases in Lemma 2. To understand the result, it is helpful to consider the two effects induced by an increase in technology quality θ. The first is a "rat race effect": Each employee anticipates that better technology will raise their coworkers' output, which creates an incentive to work more intensely to avoid falling behind and being penalized through the relative-performance component of pay. However, better technology can also induce a "survival effect": When coworker j burns out, they lose $\theta x[1 - \tau_j(x)]$ units of output, which hands employees who remain productive a relative-performance advantage. Better technology magnifies this advantage and moves employees closer to the maximum relative-performance reward, reducing the payoff from working harder. In other words, employees can benefit simply by remaining productive and avoiding burnout, which

encourages a strategic survival behavior that moderates work intensity.

The difference between parts (i) and (ii) of Lemma 2 depends on which of the two effects described above dominates. When relative-performance incentives are weak ($\gamma \leq \bar{\gamma}_R$), relative pay is too small a component of compensation for the survival effect to offset the higher return to work, so work intensity increases in θ (Lemma 2(i)). In contrast, when incentives are strong ($\gamma > \bar{\gamma}_R$) and burnout exposure ρ is sufficiently high, coworker burnout meaningfully affects each employee's work decision. The bounds on ϕ^w ensure that employees still work intensely enough for coworker burnout to be advantageous, but not so intensely that they max out the expected relative-performance pay. The net result is that the survival effect dominates at high technology levels, and work intensity decreases in θ (Lemma 2(ii)). In both cases, equilibrium mean burnout time $\mathbb{E}[\tau_i(x^*)]$ moves in the opposite direction to x^* because $\mathbb{E}[\tau_i(x)]$ decreases in x at a fixed ρ (cf. Lemma 3 in the appendix).

Lemma 2 may help explain recent evidence that AI is associated with longer working hours and more tasks (Jiang et al., 2025; Ranganathan and Ye, 2026). However, the result also suggests that the rat race effect may not continue as technology improves further. In highly competitive workplaces where burnout is common, employees may instead reduce their work intensity. Whether better technology leads to more work therefore depends on how aggressively the firm rewards differences in relative performance and how readily intensive work leads to burnout.

3.2. Impact of Technology on Employee Welfare

Equipped with Lemma 2, we can now examine how employee welfare and firm profit respond to changes in technology quality. Because better technology makes each unit of work more valuable, one might expect employees to benefit. AI assistance has indeed been shown to improve task satisfaction and lower attrition in some settings (Noy and Zhang, 2023; Brynjolfsson et al., 2025). However, whether employees are better off also depends on how much more they work to produce the additional output. Our first main result extends Lemma 2 to characterize how employee welfare is impacted by technology quality and the resulting strategic work decisions of the team:

Proposition 1 (Technology and employee welfare)**.** *Let* $\bar{\gamma}_W \equiv n(\alpha-1)/(n-1)$*. The following statements hold.*

(i) In the absence of burnout ($\rho = 0$)*, employee welfare* W^* *is strictly increasing, constant, or strictly decreasing in* θ *on* $[1, \theta^m]$ *according as* $\gamma < \bar{\gamma}_W$*,* $\gamma = \bar{\gamma}_W$*, or* $\gamma > \bar{\gamma}_W$*.*

(ii) For any $\gamma > \bar{\gamma}_W$ *and* $\phi^w > 0$*, sufficiently large* ρ *implies that* W^* *strictly decreases in* θ *on* $[1, \theta_1]$ *for some* $\theta_1 \in (1, \theta^m)$*.*

(iii) Fix $\gamma > \bar{\gamma}_W$ *and a sufficiently small* $\phi^w > 0$*. For sufficiently large* ρ*, there exist a finite* $\theta^m > 1$ *and thresholds* $1 < \theta_1 < \theta_2 < \theta^m$ *with the following property:* W^* *strictly decreases in* θ *on* $[1, \theta_1]$*. On* $[\theta_2, \theta^m]$*, equilibrium work intensity* x^* *strictly decreases in* θ*, and mean burnout time* $\mathbb{E}[\tau_i(x^*)]$ *and employee welfare* W^* *strictly increase in* θ*. Consequently,* W^* *has at least one interior minimum over* $[1, \theta^m]$*.*

Proposition 1(i) shows that in the absence of burnout, strong relative-performance incentives alone can make improvements in technology quality harmful to employee welfare. The intuition is that all employees produce the exact same realized output, so no employee receives relative-performance pay, despite the additional work induced by competition.

To understand parts (ii) and (iii), let $W(x, \theta)$ denote total employee welfare when all employees choose the same work intensity x, holding ρ and γ fixed. Because equilibrium welfare is $W^* = W(x^*, \theta)$, the total effect of better technology on welfare can be decomposed as

$$\frac{dW^*}{d\theta} = \underbrace{\frac{\partial W}{\partial \theta}}_{\text{Direct technology effect}} + \underbrace{\frac{\partial W}{\partial x}\frac{dx^*}{d\theta}}_{\text{Strategic work-intensity effect}}. \tag{5}$$

Both partial derivatives are evaluated at $x = x^*$. Under the conditions in parts (ii) and (iii), the direct technology effect is always positive, so the direction of the total effect depends on the strategic work-intensity effect. When burnout exposure is high and technology quality is low, work intensity is low and coworker burnout is rare. In this regime, the rat race effect described following Lemma 2 dominates, $dx^*/d\theta > 0$. This additional work lowers welfare, $\partial W/\partial x < 0$, because employees end up working more intensely than they would collectively prefer. The strong incentive $\gamma > \bar{\gamma}_W$ ensures that this strategic effect outweighs the direct benefit of better technology (Proposition 1(ii)).

In part (iii), a sufficiently small employee burnout cost ϕ^w allows work intensity – and hence coworker burnout – to rise fast enough for the survival effect to dominate at higher technology levels. Competition then cools and work intensity falls, $dx^*/d\theta < 0$. Both terms in (5) are then positive, and welfare recovers (Proposition 1(iii)). Note that falling intensity is sufficient but not necessary for this recovery:

Welfare can increase while intensity is still rising, provided competition has cooled enough for the direct effect to dominate. Figure 3 depicts a numerical example in which welfare begins to recover near $\theta = 3.3$, even as mean burnout time continues to fall.

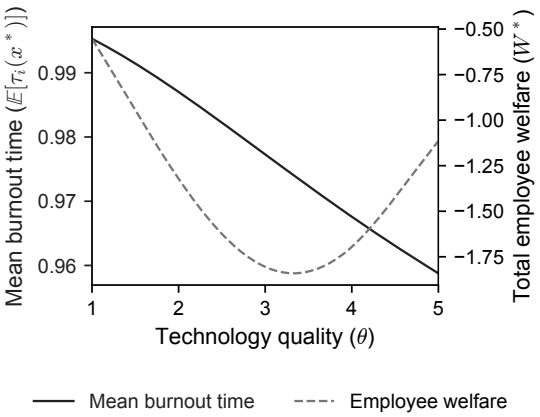


Figure 3: Employee outcomes under varying technology quality and a fixed incentive. Dashed line (right axis) illustrates the welfare decline and rebound described in Proposition 1(ii)–(iii); solid line (left axis) shows that employees burn out sooner as technology improves. Parameters: $\alpha = 3$, $n = 4$, $\gamma = 3$, $\rho = 0.01$, $\phi^w = 2$.

What does Proposition 1 imply for firms in which employees compete for rewards, such as through forced-ranking systems? As technology progresses, such firms may see output rise while employee welfare falls, even if they leave their formal reward structure unchanged. Each employee works harder to keep up with more productive coworkers, even though all employees would benefit from collectively moderating work. Employees' willingness to take on more work with AI (Ranganathan and Ye, 2026) may therefore reflect competitive pressure rather than productivity gains alone. Nonetheless, if the risk of burnout is strong enough to significantly discipline employee competition, employees may still benefit as technology improves due to the productivity gains.

3.3. Impact of Technology on Firm Profit

We now consider how technological progress affects firm profit. Because better technology both raises the value of completed work and induces more of it (Lemma 2), it is natural to expect the firm's marginal return to technology to rise as technology improves. In that case, the firm would enjoy increasing marginal returns to technology. Our next result shows that this intuition indeed holds in the absence of burnout, but that burnout risk can eventually reverse it.

Proposition 2 (Burnout and marginal returns to technology)**.** *Let $\bar{\gamma}_R = n/(n-1)$, as defined in Lemma 2. The following statements hold.*

(i) In the absence of burnout ($\rho = 0$), the marginal return to technology, $(d/d\theta)\Pi^*$, is positive and strictly increases in θ on $[1, \theta^m]$.

(ii) For any $\gamma > 0$ and $\phi^w > 0$, sufficiently large ρ implies that, for sufficiently large v, the marginal return to technology is positive and strictly increases in θ on $[1, \theta_1]$ for some $\theta_1 \in (1, \theta^m)$.

(iii) Fix $\gamma > \bar{\gamma}_R$ and a sufficiently small $\phi^w > 0$. For sufficiently large ρ, there exist a finite $\theta^m > 1$ and thresholds $1 < \theta_1 < \theta_2 < \theta^m$ with the following property: For sufficiently large v, the marginal return to technology is positive and strictly increases in θ on $[1, \theta_1]$, while it is positive and strictly decreases in θ on $[\theta_2, \theta^m]$. Consequently, $(d/d\theta)\Pi^*$ has at least one interior maximum over $[1, \theta^m]$.

Proposition 2 shows that the possibility of employee burnout can drive decreasing returns to technology for the firm. To see why, consider how the net value of employee output changes as θ increases. As in (5), this change consists of a direct effect, as each unit of work becomes more productive, and a strategic effect, as employees adjust their work intensity $x^*(\theta)$ and thereby change expected total work. The firm's marginal return to technology depends on both effects.

In the absence of burnout, better technology always intensifies work, in addition to increasing the value of work, and so the marginal return rises with technology quality (Proposition 2(i)). With burnout, marginal returns can also increase initially. As in Proposition 1(ii), when technology quality is low, high burnout exposure keeps work intensity low. In this setting, there is minimal risk of burnout, so better technology encourages additional work. The marginal return then rises when the value of output is sufficiently high (Proposition 2(ii)).

We now address part (iii). At higher technology quality, strong relative-performance incentives can make work intensity cool off, as shown in Lemma 2(ii). The condition that employee burnout cost ϕ^w is small allows work intensity to reach a level at which this cooling effect is significant. As work intensity levels off and begins to decline, expected output continues to rise, but at a decreasing rate. As a consequence, improvements to technology increase profit by progressively smaller amounts (Proposition 2(iii)). Figure 4 depicts a numerical example. Note that the marginal return peaks before work intensity: A slowdown in the growth of work intensity is enough to diminish the firm's returns. The large-v condition ensures that firm profit is sufficiently sensitive to changes in employee output.

Proposition 2 cautions against extrapolating from early gains: In competitive workplaces where employees face burnout risk, further improvements in technology may yield progressively smaller gains. Importantly, these diminishing returns need not reflect limits of the technology itself; instead, they

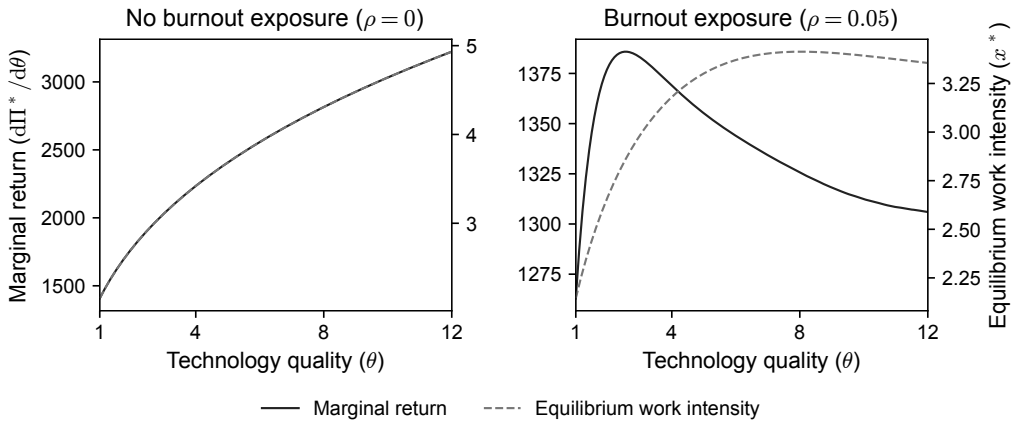


Figure 4: Marginal returns to technology with and without burnout exposure. Solid lines (left axes) depict behavior described in parts (i) and (ii)–(iii) of Proposition 2 in the left and right panels, respectively. Dashed lines (right axes) show equilibrium work intensity. Parameters: $\alpha = 4$, $n = 10$, $\gamma = 10$, $v = 50$, $\phi^w = \phi^f = 0.1$.

may arise because employees moderate their work intensity as technology improves. This behavior is consistent with evidence that contest leaders reduce effort as their lead grows (Casas-Arce and Martínez-Jerez, 2009).

To fully reap the benefits of technological progress, employers should therefore consider how better technology changes competition among employees, especially when relative-performance incentives are strong and burnout risk is high. Our results show that a firm's returns to technology can depend on both workplace competition and employee work conditions, supporting the broader view that organizational practices shape the returns to technology (Brynjolfsson and Hitt, 2000).

4. Endogenous Performance Incentives

Section 3 shows that productivity-enhancing technology can intensify competition and reduce employee welfare. In practice, these effects of improving technology might be offset by improvements in employee work conditions, such as greater flexibility over hours (Moen et al., 2016; Kelly and Moen, 2020).

In our model, an improvement in work conditions is represented by a reduction in burnout exposure ρ. Clearly, a reduction in ρ benefits employees by reducing the burnout hazard. What is less obvious is whether employees continue to benefit if the firm can redesign the incentive γ in response. We address that question next.

We treat changes in ρ as exogenous, which reflects that such improvements are often determined outside the firm, for example by regulation or cultural changes in workplace norms. For a given ρ, the firm chooses incentive strength γ to maximize equilibrium profit: $\gamma^* \in \operatorname{argmax}_{\gamma \geq 0} \Pi^*(\gamma)$. Proposition 3 characterizes the effect of an improvement in work conditions when the firm can freely adjust the

incentive γ.

Proposition 3 (Burnout exposure and the firm's optimal incentive)**.** *Fix θ and $\phi^f > 0$. The following statements hold.*

(i) Suppose the incentive $\gamma > 0$ is fixed. Then for sufficiently small $\rho > 0$, a decrease in ρ implies that equilibrium work intensity x^* and mean burnout time $\mathbb{E}[\tau_i(x^*)]$ both strictly increase.

(ii) Suppose the firm chooses the incentive γ. Then for sufficiently small $\rho > 0$, a decrease in ρ implies that the optimal incentive γ^* and work intensity x^* strictly increase, and mean burnout time $\mathbb{E}[\tau_i(x^*)]$ and employee welfare W^* strictly decrease.

(iii) Suppose $\frac{(v-1)\phi^w}{\alpha} < \phi^f < (v-1)\phi^w$ and the firm chooses the incentive γ. Then for sufficiently large ρ, a decrease in ρ implies that work intensity x^* and employee welfare W^* strictly increase, and mean burnout time $\mathbb{E}[\tau_i(x^*)]$ strictly decreases.

Part (i) establishes a benchmark with a fixed incentive: Lower burnout exposure encourages employees to work more, but when exposure is already low, the protection offered by better work conditions is strong enough that mean burnout time still increases. Part (ii) shows that this benefit does not occur if the firm can choose its incentive. Instead, the firm increases γ, and the additional work it induces outweighs the improvement in work conditions from the decrease in ρ. As a consequence, equilibrium mean burnout time falls instead. In other words, an improvement in work conditions makes burnout *more* likely if the incentive is endogenous.

To understand why this occurs, it is helpful to consider the two effects of a reduction in burnout exposure ρ. First, at fixed work intensity, lower exposure reduces the burnout hazard and keeps employees productive for longer. However, safer work conditions make the firm's productive capacity more secure, giving the firm room to demand more from its employees. As a consequence, the firm finds it profitable to increase the incentive γ^*, which ratchets up competition and induces a higher work intensity x^*. Whether mean burnout time falls thus depends on whether work intensity increases by more than the factor needed to offset lower exposure. When burnout exposure ρ is small, this factor is exceeded, so mean burnout time falls.

Why does the firm "overshoot" on work intensity and induce higher burnout risk when work conditions improve? The firm weighs the return to additional work against its marginal costs, including burnout costs. Better work conditions make additional work less costly, encouraging the firm to strengthen incentives. Employees' burnout risk, however, depends on their *cumulative* exposure over the period. When ρ is small, as in part (ii), the resulting increase in work intensity more than offsets the improvement in work conditions, leaving employees more likely to burn out.

Part (iii) returns to a setting where employees benefit from an improvement in work conditions, despite the firm ratcheting up competition by increasing γ. In particular, at sufficiently high exposure, the condition in part (iii) places the employer's burnout cost in an intermediate range: low enough that the firm chooses a strictly positive relative-performance incentive $\gamma^* > 0$, but high enough to limit the work intensity it induces. This balance ensures that improved work conditions increase expected pay enough to outweigh the additional expected work and burnout costs. A decrease in mean burnout time can therefore accompany either a welfare gain or a welfare loss. The left panel of Figure 5 depicts both cases.

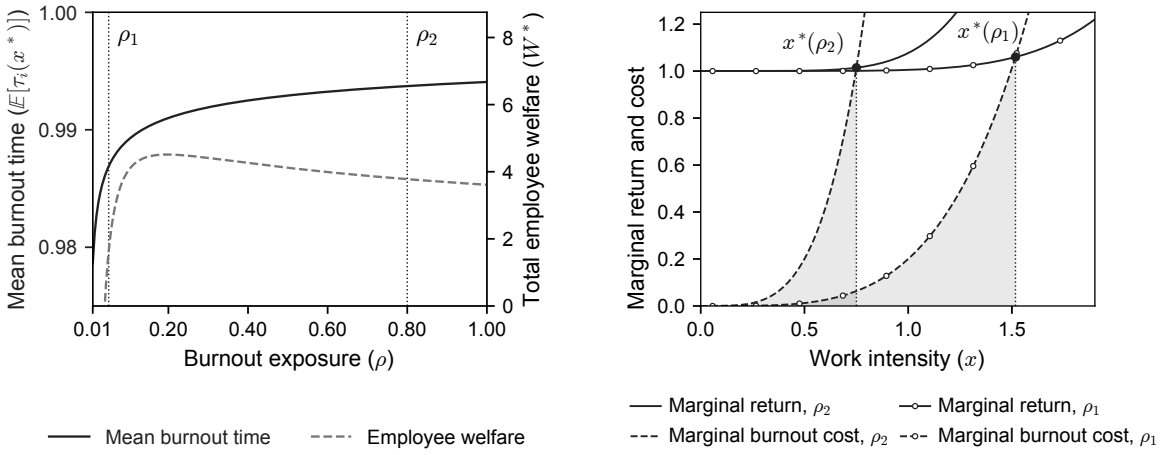


Figure 5: Burnout exposure and employee outcomes under the firm's optimal incentive. *Left panel*: The solid line (left axis) depicts the mean burnout time behavior in Proposition 3(ii)–(iii). The dashed line depicts the employee welfare behavior in parts (ii) and (iii). *Right panel*: The firm's marginal return (solid) and marginal burnout cost (dashed) intersect at its implemented work intensity. Shaded areas are proportional to the cumulative burnout hazard under each exposure level; larger areas thus correspond to earlier burnout in expectation. Parameters: $\alpha = 5$, $n = 4$, $\theta = 2$, $v = 1.5$, $\phi^w = 13.35$, $\phi^f = 4$; $\rho_1 = 0.05$ (circles) and $\rho_2 = 0.8$ (no markers).

We now revisit the role of technology. Corollary 1 shows that the firm's control over the incentive also changes how technology affects employees:

Corollary 1 (Technology under endogenous incentives)**.** *Fix $\phi^f > 0$. When the firm chooses the incentive γ, for sufficiently small $\rho > 0$, equilibrium work intensity x^* strictly increases and mean burnout time $\mathbb{E}[\tau_i(x^*)]$ strictly decreases in technology quality θ on $[1, \theta^m]$.*

When exposure is sufficiently low, work intensity continues to rise and mean burnout time falls under the firm's optimal incentive as technology improves. The decline in work intensity and increase in mean burnout time that can arise under fixed incentives in Section 3 are thus absent in this low-exposure region when the incentive is endogenous.

Proposition 3 adds an important caveat to the notion that improved work conditions can buffer

employees against burnout (Bakker et al., 2005): Not only can the firm "optimize away" these protections by adjusting its performance incentives, but employees can be left worse off overall under the improved work conditions because of how aggressively the firm responds. For employees, the benefit of greater schedule flexibility or other preventive support therefore depends on whether the firm also raises performance expectations in tandem. Corollary 1 further suggests that an improvement in technology quality can drive the firm to create a more competitive environment, increasing burnout. These results suggest that any evaluation of new policies or norms that improve work conditions should account for the firm's incentive response: Their direct benefits to employees must be weighed against potentially increased pressure to perform.

5. Mitigation and Social Welfare

Investments in employee well-being, including wellness programs, provide a natural starting point for assessing how well a workplace protects its employees. However, Section 4 shows that work conditions and performance incentives are closely linked: When the work environment improves, a profit-maximizing firm may respond by strengthening its incentive and inducing even more work.

We now consider investment in burnout mitigation jointly with the choice of performance incentive. We compare a profit-maximizing firm with one that maximizes social welfare – the sum of firm profit and employee welfare. This benchmark reflects the choices of a regulator, or of a firm that fully internalizes employees' costs.

Formally, we allow the firm to choose the burnout exposure parameter ρ, a stylized representation of policy choices or investments in burnout mitigation, such as greater scheduling flexibility or wellness programs. Let $C(\rho)$ denote the per-employee cost required to achieve ρ. The firm chooses ρ jointly with the performance incentive γ, after which employees choose their work intensities.

Assumption 1 (Burnout mitigation cost)**.** *Mitigation spending per employee is $C(\rho) > 0$, where C is continuously differentiable with $(\partial/\partial\rho)C(\rho) < 0$, $\lim_{\rho\to 0} \rho C(\rho) = \infty$, and $\lim_{\rho\to\infty} \rho C(\rho) = 0$.*

Assumption 1 imposes regularity conditions on $C(\rho)$ such that it is optimal for the firm to choose a strictly positive and finite value of ρ.

We consider both profit-maximizing and social welfare-maximizing objectives. Using the equilibrium profit Π^* obtained from (3), the profit-maximizing design $(\rho^F(\theta), \gamma^F(\theta))$ maximizes $\Pi^*(\rho, \gamma) - nC(\rho)$ over $\rho > 0$ and $\gamma \geq 0$. For social welfare, combining firm profit in (3) with the sum of employee utilities in (2) and subtracting mitigation spending gives the social welfare objective

$$\Pi^*(\rho, \gamma) + W^*(\rho, \gamma) - nC(\rho). \tag{6}$$

Let $(\rho^S(\theta), \gamma^S(\theta))$ denote any design that maximizes (6), and let τ_i^S and τ_i^F denote employee i's censored burnout times under the social welfare-maximizing and profit-maximizing designs, each evaluated at its own equilibrium intensity and exposure. Their means are $\mathbb{E}[\tau_i^S]$ and $\mathbb{E}[\tau_i^F]$. Proposition 4 characterizes the optimal choice of ρ and γ under both the profit-maximizing and social welfare-maximizing objectives, and the implications for employees' equilibrium mean burnout times.

Proposition 4 (Social welfare and burnout mitigation)**.** *Suppose Assumption 1 holds. There exists $\underline{\theta} > 1$ such that, for sufficiently large ϕ^w and sufficiently small $\phi^f > 0$, the following comparisons hold for all $\theta \in [1, \underline{\theta}]$. Relative to the profit-maximizing design, the social welfare-maximizing design has lower mitigation spending, $C(\rho^S) < C(\rho^F)$; higher burnout exposure, $\rho^S > \rho^F$; a weaker incentive, $\gamma^S < \gamma^F$; and a higher mean burnout time, $\mathbb{E}[\tau_i^S] > \mathbb{E}[\tau_i^F]$.*

Proposition 4 gives sufficient conditions for the comparison when both choices are endogenous: With large employee burnout costs, small employer burnout costs, and low technology quality, the social welfare-maximizing design spends less on mitigation while achieving higher mean burnout time.

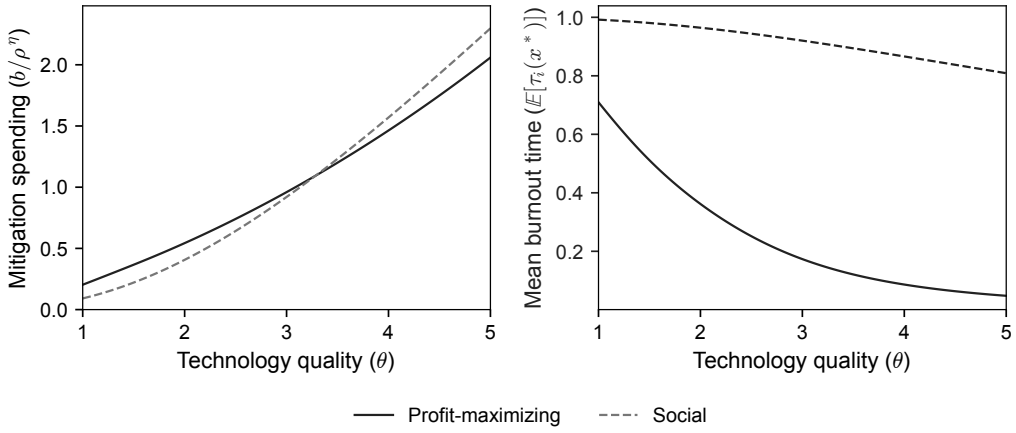


Figure 6: Comparison of profit-maximizing (solid) and social welfare-maximizing (dashed) workplace designs. *Left panel*: At low technology quality, the social welfare-maximizing firm spends less on mitigation, illustrating Proposition 4. *Right panel*: The profit-maximizing firm induces faster burnout (i.e., shorter mean burnout time) at low technology quality, illustrating Proposition 4. Parameters: $\alpha = 3$, $n = 4$, $v = 3$, $\phi^w = 4$, $\phi^f = 0.25$, $b = 20$, $\eta = 1.5$; mitigation cost $C(\rho) = b/\rho^\eta$ satisfies Assumption 1.

The result in Proposition 4 is driven by the different ways in which the two designs account for burnout. In the profit function Π^*, burnout affects expected total work, employee payments, and the employer burnout cost $n\phi^f \Pr(\tau_i(x^*) < 1)$. In contrast, the social welfare objective in (6) also counts employees' active-work cost, $n\rho^{-1} \Pr(\tau_i(x^*) < 1)$, and burnout loss, $n\phi^w \Pr(\tau_i(x^*) < 1)$. When ϕ^w is high and ϕ^f is low, the social design therefore has a stronger reason to reduce work, which it does by weakening the incentive ($\gamma^S < \gamma^F$).

Burnout mitigation is valuable only to the extent that there is work to protect. Because the social

welfare design induces weaker competition (lower γ), and thus demands less work from employees, it does not need to spend as much on mitigation. When technology quality is low, the output sacrificed by reducing work is small, so the social welfare design prefers to reduce work in the first place rather than protect it ($\rho^S > \rho^F$). Although the higher exposure induced by the social welfare design raises the burnout hazard at a given work intensity, the reduction in work intensity is large enough that mean burnout time remains higher under the social welfare design, $\mathbb{E}[\tau_i^S] > \mathbb{E}[\tau_i^F]$. In summary, the incentive determines how much work is performed, and mitigation protects that work – a firm that internalizes employee welfare uses less of both in the parameter region considered.

In the numerical example in Figure 6, the profit-maximizing design induces a much lower mean burnout time once technology quality is sufficiently high (with parameters chosen to satisfy Proposition 4). In this example, the firm extracts the maximum work each employee can sustain and simply absorbs burnout as the cost of doing so. At higher technology quality, however, reducing work sacrifices more valuable output, and mitigation can become the better mechanism for protecting employees, which is why the social welfare design eventually spends more on mitigation when θ is large.

In practice, this result suggests that an employer's spending on wellness and burnout mitigation can be a misleading measure of how well it protects employees. Under the conditions in Proposition 4, a profit-maximizing firm chooses to spend more on mitigation only because it induces more intensive work through strong incentives, and has more productivity to protect from burnout. A firm's spending on employee well-being may therefore reflect the demands it places on employees in addition to its valuation of employee well-being.

6. Discussion

This paper aims to provide theoretical grounding for early empirical evidence suggesting AI might intensify work (Jiang et al., 2025; Ranganathan and Ye, 2026). Our mechanism is grounded in workplace competition: When employee rewards depend strongly on relative performance, better technology can induce employees to work harder to keep pace with anticipated gains in coworkers' productivity, leaving employees collectively worse off.

Performance management is increasingly a focus of organizations (Cappelli and Tavis, 2016), and prior research has documented adverse effects of performance rankings, including reduced cooperation and information sharing (Chambers and Baker, 2020; Rao, 2025). Our analysis suggests better technology might amplify some of the undesirable effects of performance rankings, like burnout (Proposition 1). The model also generates new predictions that can be tested empirically: Burnout-related outcomes like absences or medical leaves should be more pronounced in organizations with strong

relative-performance incentives.

For employers promoting AI adoption, accounting for potentially adverse welfare effects may mean loosening forced-ranking quotas or placing more weight on other measures of performance, such as collaboration. PayPal, for example, formally removed individual performance ratings in 2021 with the stated aim of reducing employee stress (PayPal, 2022). Our analysis suggests performance evaluation systems may continue to attract scrutiny in the near future if technology indeed amplifies burnout.

Employee incentives may also impact firms' returns to technology. In high-pressure environments where employee well-being is at risk, further improvements in technology may yield progressively smaller gains if employees strategically moderate their workloads (Proposition 2). Crucially, this can occur not only because employees seek to protect themselves, but also because they anticipate burnout-driven declines in their coworkers' productivity. This offers an additional justification for investing in employee well-being – to fully reap the benefits of technological progress.

Policies that improve work conditions, such as greater schedule flexibility, may reduce burnout (Moen et al., 2016). However, if employers respond by raising performance expectations, the additional work they demand can more than offset this protection, lowering employee welfare (Proposition 3). Kelly and Moen (2020) argue that reducing overload requires work redesign, including giving employees more control over how they work and clear direction on their goals. Our results add that employers should commit to holding performance incentives fixed when work conditions improve, so that the better work conditions materially translate into gains to well-being.

Our results also suggest that ongoing debates over policies such as right-to-disconnect laws and four-day workweeks (Fair Work Ombudsman, 2026; Fan et al., 2025) should consider how these policies interact with firms' performance incentives and with the productivity gains from AI. Genuine limits on effort could help curb burnout, whereas limits on total working hours may leave room for more intensive work within those hours. Although we do not model these policies directly, Proposition 3(ii) identifies a potential risk: If they reduce burnout exposure without meaningfully constraining employee effort, firms may respond by intensifying competition to the point of reversing the welfare gains.

Spending on wellness programs may also provide an incomplete picture of employee protection: An employer that demands intensive work has more productivity to defend, and so may spend more on burnout prevention than a welfare-conscious employer (Proposition 4). Our results therefore suggest evaluating mitigation spending alongside the competitive pressure created by performance incentives. Croft et al. (2024) attribute the muted impact of wellness programs to their focus on individual employees rather than broader work conditions. Our results support this view: When the productivity gains from technology are modest, moderating competition may be a more welfare-efficient approach

to protecting employees than additional wellness spending.

Our model omits several features of real workplaces. Employees may differ in how effectively they use technology, one employee's burnout may shift work onto coworkers, and work intensity may change over time. We also do not account for employees potentially quitting, which has been associated with forced-ranking systems (Bond, 2026). Incorporating these features may change our results or generate new ones. Firms may also use other levers, such as team-based rewards, which can dampen competition but introduce other challenges like free-riding. We also assume technology quality is exogenous; considering the firm's technology adoption decision and how it interacts with its performance evaluation system is a natural direction for future work.

References

ActivTrak Productivity Lab (2026). 2026 State of the workplace: The rise of amplified work. Technical report, ActivTrak.

Adobe (2026). Wellbeing reimbursement program. Adobe Benefits.

Altman, S. (2026). Sam Altman—How to start a startup. Interview by Ti Morse, Relentless podcast. July 25; relevant discussion at 45:10–46:36.

Bakker, A. B., Demerouti, E., and Euwema, M. C. (2005). Job resources buffer the impact of job demands on burnout. *Journal of Occupational Health Psychology*, 10(2):170–180.

Bakshi, N. and Peura, H. (2022). Prevent or report? Managing near misses for safer operations. *Manufacturing & Service Operations Management*, 24(4):2064–2080.

Bastani, H. and Cachon, G. P. (2025). The human-AI contracting paradox. Working paper.

Bavafa, H. and Jónasson, J. O. (2024). The distributional impact of fatigue on performance. *Management Science*, 70(5):3319–3337.

Berry Jaeker, J. A. and Tucker, A. L. (2017). Past the point of speeding up: The negative effects of workload saturation on efficiency and patient severity. *Management Science*, 63(4):1042–1062.

Bond, B. M. (2026). Cut to the curve: Underrecognition and talent loss from forced ranking in a multinational firm. *Management Science*, 72(4):3487–3507.

Boudreau, J., Hopp, W. J., McClain, J. O., and Thomas, L. J. (2003). On the interface between operations and human resources management. *Manufacturing & Service Operations Management*, 5(3):179–202.

Boyacı, T., Canyakmaz, C., and de Véricourt, F. (2024). Human and machine: The impact of machine input on decision making under cognitive limitations. *Management Science*, 70(2):1258–1275.

Boyacı, T., Canyakmaz, C., and de Véricourt, F. (2026). Beyond the black box: Unraveling the role of explainability in human-artificial intelligence collaboration. *Management Science*. Articles in Advance.

Brynjolfsson, E. and Hitt, L. M. (2000). Beyond computation: Information technology, organizational transformation and business performance. *Journal of Economic Perspectives*, 14(4):23–48.

Brynjolfsson, E., Li, D., and Raymond, L. R. (2025). Generative AI at work. *Quarterly Journal of Economics*, 140(2):889–942.

Cappelli, P. and Tavis, A. (2016). The performance management revolution. *Harvard Business Review*, 94(10).

Casas-Arce, P. and Martínez-Jerez, F. A. (2009). Relative performance compensation, contests, and dynamic incentives. *Management Science*, 55(8):1306–1320.

Chambers, C. R. and Baker, W. E. (2020). Robust systems of cooperation in the presence of rankings: How displaying prosocial contributions can offset the disruptive effects of performance rankings. *Organization Science*, 31(2):287–307.

Connelly, B. L., Tihanyi, L., Crook, T. R., and Gangloff, K. A. (2014). Tournament theory: Thirty years of contests and competitions. *Journal of Management*, 40(1):16–47.

Corbeanu, A., Iliescu, D., Ion, A., and Spînu, R. (2023). The link between burnout and job performance: A meta-analysis. *European Journal of Work and Organizational Psychology*, 32(4):599–616.

Corbett, C. J. (2024). OM Forum—The operations of well-being: An operational take on happiness, equity, and sustainability. *Manufacturing & Service Operations Management*, 26(2):409–430.

Cordes, C. L. and Dougherty, T. W. (1993). A review and an integration of research on job burnout. *Academy of Management Review*, 18(4):621–656.

Croft, J., Parks, A., and Whillans, A. (2024). Why workplace well-being programs don't achieve better outcomes. Harvard Business Review (online), October 18.

Cui, Z., Demirer, M., Jaffe, S., Musolff, L., Peng, S., and Salz, T. (2025). The effects of generative AI on high-skilled work: Evidence from three field experiments with software developers. Preprint, Microsoft Research.

Dai, T. and Taylor, T. A. (2026). Designing enterprise AI systems: Hallucination, creativity, and moral hazard. Working paper, April 6.

Dell'Acqua, F., McFowland, III, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K. C., Rajendran, S., Krayer, L., Candelon, F., and Lakhani, K. R. (2023). Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality. Working Paper 24-013, Harvard Business School.

Dixit, P. and Mann, J. (2025). Meta is about to start rating more workers as "below expectations," internal memo shows. *Business Insider*.

Dizdarer, T., Mitrofanov, D., and Reyes, R. (2025). Will AI trigger layoffs and kill the gig? The role of generalist AI under hybrid workforce. Working paper, revised August 24, 2026.

Eichenwald, K. (2012). Microsoft's lost decade. *Vanity Fair*.

Fair Work Ombudsman (2026). Right to disconnect. Updated June 18, 2026. Accessed October 29, 2026.

Fan, W., Schor, J. B., Kelly, O., and Gu, G. (2025). Work time reduction via a 4-day workweek finds improvements in workers' well-being. *Nature Human Behaviour*, 9:2153–2168.

Fang, Z., Chang, Y., Luo, X., Wu, Q., and Aspara, J. (2026). Artificial intelligence, emotional labor, and service operations. *Manufacturing & Service Operations Management*. Articles in Advance.

Feldman, P., Segev, E., and Sekar, S. (2026). Pricing human-AI services: The role of observability and congestion. Working paper, May 20.

Fisher, J., Gilmartin, M., and Silverglate, P. H. (2023). As workforce well-being dips, leaders ask: What will it take to move the needle? *Deloitte Insights*.

Fügener, A., Walzner, D. D., and Gupta, A. (2026). Roles of artificial intelligence in collaboration with humans: Automation, augmentation, and the future of work. *Management Science*, 72(1):538–557.

Gans, J. S. and Goldfarb, A. (2025). O-ring automation. Technical report, SSRN.

Giebel, M. and Lammers, A. (2025). The impact of a new workplace technology on employees. *Oxford Bulletin of Economics and Statistics*, 87(5):1003–1024.

Goh, J., Pfeffer, J., and Zenios, S. A. (2016). The relationship between workplace stressors and mortality and health costs in the United States. *Management Science*, 62(2):608–628.

Green, C. and Rahmani, M. (2022). The implications of rating systems on workforce performance. *IISE Transactions*, 54(2):159–172.

Green, J. R. and Stokey, N. L. (1983). A comparison of tournaments and contracts. *Journal of Political Economy*, 91(3):349–364.

Green, L. V., Savin, S., and Savva, N. (2013). "Nursevendor problem": Personnel staffing in the presence of endogenous absenteeism. *Management Science*, 59(10):2237–2256.

Greiner, B., Grünwald, P., Lindner, T., Lintner, G., and Wiernsperger, M. (2026). Incentives, framing, and reliance on algorithmic advice: An experimental study. *Management Science*, 72(1):302–322.

Guan, X., Qi, A., and Wang, S. (2025). Why the best machine may not be the best: Incentivizing human-machine collaboration. Working paper, revised September 30, 2026.

Gubler, T., Larkin, I., and Pierce, L. (2018). Doing well by making well: The impact of corporate wellness programs on employee productivity. *Management Science*, 64(11):4967–4987.

Habel, J., Alavi, S., and Linsenmayer, K. (2021). Variable compensation and salesperson health. *Journal of Marketing*, 85(3):130–149.

Hu, M., Jin, Y., and Nejati, E. (2026). AI co-creation. Working paper.

HubSpot (2021). 2021 Sustainability report. Technical report, HubSpot.

Hyder, B. (2022). Why Salesforce's chief people officer takes mental health so seriously. *Salesforce News & Insights*.

Jiang, W., Park, J., Xiao, R. J., and Zhang, S. (2025). AI and the extended workday: Productivity, contracting efficiency, and distribution of rents. NBER Working Paper 33536, National Bureau of Economic Research.

Jiao, C., Qi, A., and Chen, J. (2026). Combating excessive overtime in global supply chains: The workforce perspective. *Management Science*. Articles in Advance.

Jones, D., Molitor, D., and Reif, J. (2019). What do workplace wellness programs do? Evidence from the Illinois Workplace Wellness Study. *Quarterly Journal of Economics*, 134(4):1747–1791.

KC, D. S. and Terwiesch, C. (2009). Impact of workload on service time and patient safety: An econometric analysis of hospital operations. *Management Science*, 55(9):1486–1498.

Kelly, E. L. and Moen, P. (2020). Fixing the overload problem at work. *MIT Sloan Management Review*. Summer 2020 issue.

Kremer, M. (1993). The O-ring theory of economic development. *Quarterly Journal of Economics*, 108(3):551–575.

Kuntz, L., Mennicken, R., and Scholtes, S. (2015). Stress on the ward: Evidence of safety tipping points in hospitals. *Management Science*, 61(4):754–771.

Landers, R. M., Rebitzer, J. B., and Taylor, L. J. (1996). Rat race redux: Adverse selection in the determination of work hours in law firms. *American Economic Review*, 86(3):329–348.

Lazear, E. P. (1989). Pay equality and industrial politics. *Journal of Political Economy*, 97(3):561–580.

Lazear, E. P. (2000). Performance pay and productivity. *American Economic Review*, 90(5):1346–1361.

Lazear, E. P. and Rosen, S. (1981). Rank-order tournaments as optimum labor contracts. *Journal of Political Economy*, 89(5):841–864.

Legros, B. and de Véricourt, F. (2026). Human-AI interaction in congested service systems: When it improves performance, and when it doesn't. Working paper, February 3.

Liu, H., Ding, N., Li, X., Chen, Y., Sun, H., Huang, Y., Liu, C., Ye, P., Jin, Z., Bao, H., and Xue, H. (2024). Artificial intelligence and radiologist burnout. *JAMA Network Open*, 7(11):e2448714.

Lu, T. and Tomlin, B. (2026). Augmenting the operations manager with a prediction machine. Working paper, April 23.

Martinez, M. F., O'Shea, K. J., Kern, M. C., Chin, K. L., Dinh, J. V., Bartsch, S. M., Weatherwax, C., Velmurugan, K., Heneghan, J. L., Moran, T. H., Scannell, S. A., John, D. C., Shah, T. D., Petruccelli, S. A., White, C., Dibbs, A. M., and Lee, B. Y. (2025). The health and economic burden of employee burnout to U.S. employers. *American Journal of Preventive Medicine*, 68(4):645–655.

Maslach, C., Schaufeli, W. B., and Leiter, M. P. (2001). Job burnout. *Annual Review of Psychology*, 52:397–422.

Matros, A., Smirnov, V., Wait, A., and Zhang, H. (2026). Technology, work intensification and burnout. *Economic Record*. Early View.

Mazmanian, M., Orlikowski, W. J., and Yates, J. (2013). The autonomy paradox: The implications of mobile email devices for knowledge professionals. *Organization Science*, 24(5):1337–1357.

Moen, P., Kelly, E. L., Fan, W., Lee, S.-R., Almeida, D. M., Kossek, E. E., and Buxton, O. M. (2016). Does a flexibility/support organizational initiative improve high-tech employees' well-being? Evidence from the Work, Family, and Health Network. *American Sociological Review*, 81(1):134–164.

Mookherjee, D. (1984). Optimal incentive schemes with many agents. *Review of Economic Studies*, 51(3):433–446.

Nalebuff, B. J. and Stiglitz, J. E. (1983). Prizes and incentives: Towards a general theory of compensation and competition. *Bell Journal of Economics*, 14(1):21–43.

Noy, S. and Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654):187–192.

PayPal (2022). 2021 global impact report. Technical report, PayPal Holdings, Inc.

Ranganathan, A. and Ye, X. M. (2026). AI doesn't reduce work—it intensifies it. *Harvard Business Review*.

Rao, K. (2025). The effects of competition and scarcity on interpersonal communication in organizations. *Organization Science*, 36(5):1939–1961.

Roels, G. and Staats, B. R. (2021). OM Forum—People-centric operations: Achievements and future research directions. *Manufacturing & Service Operations Management*, 23(4):745–757.

Salvagioni, D. A. J., Melanda, F. N., Mesas, A. E., González, A. D., Gabani, F. L., and de Andrade, S. M. (2017). Physical, psychological and occupational consequences of job burnout: A systematic review of prospective studies. *PLOS ONE*, 12(10):e0185781.

Schaufeli, W. B. and Bakker, A. B. (2004). Job demands, job resources, and their relationship with burnout and engagement: A multi-sample study. *Journal of Organizational Behavior*, 25(3):293–315.

Schaufeli, W. B., Bakker, A. B., and Van Rhenen, W. (2009). How changes in job demands and resources predict burnout, work engagement, and sickness absenteeism. *Journal of Organizational Behavior*, 30(7):893–917.

Schultz, K. L., Juran, D. C., Boudreau, J. W., McClain, J. O., and Thomas, L. J. (1998). Modeling and worker motivation in JIT production systems. *Management Science*, 44(12):1595–1607.

Siderius, J., Shumsky, R. A., and Taylor, A. (2026). Use it or slowly lose it: Expertise atrophy with organizational AI usage. Working paper, March 11.

Song, H., Tucker, A. L., Murrell, K. L., and Vinson, D. R. (2018). Closing the productivity gap: Improving worker productivity through public relative performance feedback and validation of best practices. *Management Science*, 64(6):2628–2649.

Soper, S. (2022). Amazon is focus of push to curb "rank-and-yank" worker ratings. *Los Angeles Times*.

Tan, T. F. and Netessine, S. (2014). When does the devil make work? An empirical study of the impact of workload on worker productivity. *Management Science*, 60(6):1574–1593.

Torrence, R. and Mascarenhas, N. (2026). AI is making Silicon Valley productive, anxious and afraid to log off. *Bloomberg*. June 26.

Upwork Research Institute (2024). Upwork study finds employee workloads rising despite increased C-suite investment in artificial intelligence. Press release, July 23.

Waiser, R., Dubra, J., and Benoît, J.-P. (2025). Incentives, burnout, and turnover: Dynamic compensation design with effort cost spillover. *Quantitative Marketing and Economics*, 23(4):589–639.

Weibull, W. (1951). A statistical distribution function of wide applicability. *Journal of Applied Mechanics*, 18(3):293–297.

World Health Organization (2019). Burn-out an "occupational phenomenon". ICD-11 Frequently Asked Questions.

A Proofs

Lemma 3. *For $x > 0$ and $\rho > 0$, the following statements hold.*

(i) Completed work has survival function $\Pr(x\tau_i(x) > k) = e^{-\rho k^\alpha/\alpha}$ for $0 \le k < x$ and zero for $k \ge x$. Thus,

$$\mathbb{E}[x\tau_i(x)] = \int_0^x e^{-\rho k^\alpha/\alpha}\,dk, \qquad \mathbb{E}[(x\tau_i(x))^2] = \int_0^x 2k\,e^{-\rho k^\alpha/\alpha}\,dk. \tag{7}$$

For every x, $\mathbb{E}[x\tau_i(x)] \le c_\alpha \rho^{-1/\alpha}$ and $\mathbb{E}[(x\tau_i(x))^2] \le c_\alpha \rho^{-2/\alpha}$, where c_α depends only on α.

(ii) Completed work $x\tau_i(x)$ is stochastically increasing in x, and work lost to burnout $x[1-\tau_i(x)]$ is stochastically increasing in x and ρ. Expected completed work satisfies $(\partial/\partial x)\mathbb{E}[x\tau_i(x)] = e^{-\rho x^\alpha/\alpha}$.

(iii) Expected completed work $\mathbb{E}[x\tau_i(x)]$ and mean burnout time $\mathbb{E}[\tau_i(x)]$ strictly decrease in ρ, while expected lost work $\mathbb{E}[x(1-\tau_i(x))]$ strictly increases. Also, $|\rho(\partial/\partial\rho)\mathrm{Var}(x\tau_i(x))| \le 4x^2/\alpha$.

(iv) The distribution of $\tau_i(x)$ depends on x and ρ only through $u = \rho x^\alpha/\alpha$, with $\Pr(\tau_i(x) \ge t) = e^{-ut^\alpha}$ for $t \in [0,1]$. Mean burnout time $\mathbb{E}[\tau_i(x)]$ strictly decreases in u. As $u \to 0$, $\mathbb{E}[\tau_i(x)] \to 1$ and $\mathrm{Var}(\tau_i(x)) \to 0$. As $u \to \infty$, $\mathbb{E}[\tau_i(x)] \to 0$ and $\mathrm{Var}(\tau_i(x))/\mathbb{E}[\tau_i(x)] \to 0$.

Proof. (i). The survival function follows from the burnout distribution in Section 2.2 by setting $t = k/x$ and using $\tau_i(x) \le 1$. Integrating this survival function gives (7). To obtain the bounds, extend the integrals to infinity and substitute $w = \rho^{1/\alpha}k$. Choose c_α at least as large as the finite integrals $\int_0^\infty e^{-w^\alpha/\alpha}\,dw$ and $\int_0^\infty 2we^{-w^\alpha/\alpha}\,dw$.

(ii). The survival function in part (i) weakly increases in x. Also, $\Pr(x[1-\tau_i(x)] > l) = 1 - e^{-\rho(x-l)^\alpha/\alpha}$ for $0 \le l < x$, and zero for $l \ge x$. This probability weakly increases in x and ρ, which proves the claims about lost work. Differentiating (7) gives the stated derivative of expected completed work. Using $\mathbb{E}[x(1-\tau_i(x))] = x - \mathbb{E}[x\tau_i(x)]$ also gives $(\partial/\partial x)\mathbb{E}[x(1-\tau_i(x))] = 1 - e^{-\rho x^\alpha/\alpha}$ and $(\partial/\partial x)\mathrm{Var}(x\tau_i(x)) = 2e^{-\rho x^\alpha/\alpha}\mathbb{E}[x(1-\tau_i(x))]$.

(iii). The monotonicity claims follow from (7). Differentiating its first two moments with respect to ρ and integrating by parts gives $\alpha\rho(\partial/\partial\rho)\mathbb{E}[x\tau_i(x)] = xe^{-\rho x^\alpha/\alpha} - \mathbb{E}[x\tau_i(x)]$ and $\alpha\rho(\partial/\partial\rho)\mathbb{E}[(x\tau_i(x))^2] = 2\{x^2e^{-\rho x^\alpha/\alpha} - \mathbb{E}[(x\tau_i(x))^2]\}$. Hence,

$$\alpha\rho(\partial/\partial\rho)\mathrm{Var}(x\tau_i(x)) = 2xe^{-\rho x^\alpha/\alpha}\mathbb{E}[x(1-\tau_i(x))] - 2\mathrm{Var}(x\tau_i(x)). \tag{8}$$

Because $0 \le \mathbb{E}[x(1-\tau_i(x))] \le x$ and $0 \le \mathrm{Var}(x\tau_i(x)) \le x^2$, (8) gives the stated bound.

(iv). The burnout distribution in Section 2.2 gives the stated survival function, so $\mathbb{E}[\tau_i(x)] = \int_0^1 e^{-ut^\alpha}\,dt$ and $\mathbb{E}[\tau_i(x)^2] = \int_0^1 2te^{-ut^\alpha}\,dt$. Differentiating the first integral gives $(d/du)\mathbb{E}[\tau_i(x)] = -\int_0^1 t^\alpha e^{-ut^\alpha}\,dt < 0$, establishing strict monotonicity. Both integrals tend to one as $u \to 0$. As $u \to \infty$, substituting $w = u^{1/\alpha}t$ shows that $u^{1/\alpha}\mathbb{E}[\tau_i(x)]$ and $u^{2/\alpha}\mathbb{E}[\tau_i(x)^2]$ converge to positive constants. The remaining limits follow. □

Proof of Lemma 1. Fix $\mathbf{x}_{-i}$ and condition on coworkers' completed work $\{x_j\tau_j(x_j) : j \neq i\}$. By Lemma 3(i), employee i's completed work $x_i\tau_i(x_i)$ has density $\rho k^{\alpha-1}e^{-\rho k^\alpha/\alpha}$ on $(0, x_i)$ and mass $e^{-\rho x_i^\alpha/\alpha}$ at x_i. Differentiating expected compensation $\mathbb{E}[y_i + r_i]$ cancels the boundary terms, leaving $e^{-\rho x_i^\alpha/\alpha}$ times marginal compensation at completed work x_i. Evaluating the expected work-cost integral in (2) gives $(1 - e^{-\rho x_i^\alpha/\alpha})/\rho$. Using $(\partial/\partial y_i)r_i = \frac{n-1}{n}[\gamma - (y_i - \bar{y})]_+$ and differentiating the work and burnout cost terms in (2) gives

$$(\partial/\partial x_i)u_i = e^{-\rho x_i^\alpha/\alpha}\left\{\theta\left[1 + \tfrac{n-1}{n}\mathbb{E}\left[\left(\gamma - \tfrac{n-1}{n}\theta x_i + \frac{\theta}{n}\sum_{j\neq i} x_j\tau_j(x_j)\right)_+\right]\right] - (1+\rho\phi^w)x_i^{\alpha-1}\right\}. \tag{9}$$

The term in braces is positive at $x_i = 0$ and strictly decreases to $-\infty$: expected marginal compensation weakly decreases in x_i, while $(1+\rho\phi^w)x_i^{\alpha-1}$ strictly increases to infinity. Since $e^{-\rho x_i^\alpha/\alpha} > 0$, each employee has a unique positive best response, given by $(\partial/\partial x_i)u_i = 0$.

Setting all intensities equal to x in the first-order condition (9) gives (4). Its left-hand side strictly increases from zero to infinity in x. Its right-hand side is positive, bounded, and weakly decreasing, by Lemma 3. Thus, there is a unique positive solution x^* and a unique symmetric equilibrium.

To rule out asymmetric equilibria, suppose an equilibrium has $x_i > x_j$, and write $S = \sum_{k\notin\{i,j\}} x_k\tau_k(x_k)$ for the completed work of their remaining coworkers. By Lemma 3(ii), $x_j\tau_j(x_j)$ is stochastically no greater than $x_i\tau_i(x_i)$, and both are independent of S. Thus, since $x_i > x_j$ and $(\cdot)_+$ is weakly increasing,

$$\mathbb{E}\left[\left(\gamma - \tfrac{n-1}{n}\theta x_i + \tfrac{\theta}{n}\{x_j\tau_j(x_j) + S\}\right)_+\right] \le \mathbb{E}\left[\left(\gamma - \tfrac{n-1}{n}\theta x_j + \tfrac{\theta}{n}\{x_i\tau_i(x_i) + S\}\right)_+\right].$$

The first-order conditions in (9) therefore imply $(1+\rho\phi^w)x_i^{\alpha-1} \le (1+\rho\phi^w)x_j^{\alpha-1}$, contradicting $x_i > x_j$. Every equilibrium is therefore symmetric and, by the preceding construction, unique. □

If $\gamma \ge \frac{n-1}{n}\theta x$, the expression inside $(\cdot)_+$ in (4) is nonnegative for every realization of coworkers' lost work. Taking expectations then gives

$$(1+\rho\phi^w)x^{\alpha-1} = \theta\left[1 + \frac{n-1}{n}\gamma - \left(\frac{n-1}{n}\right)^2\theta\,\mathbb{E}[x(1-\tau_i(x))]\right]. \tag{10}$$

Because lost work $x[1-\tau_j(x)]$ equals 0 with positive probability and has a density on $(0, x)$, the sum $\frac{\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)]$ equals $\gamma > 0$ with probability zero. The expectation in (4) is therefore continuously differentiable in x, θ,

ρ, and γ. At $x = x^* > 0$, the left-hand side of (4) has a positive derivative with respect to x, while the right-hand side has a nonpositive derivative. The implicit function theorem then gives continuous differentiability of x^*, W^*, and Π^* in θ, ρ, γ, and ϕ^w. When $\gamma > \frac{n-1}{n}\theta x^*$, x^* is infinitely differentiable by (10), because $\mathbb{E}[x(1-\tau_i(x))] = x - \int_0^x e^{-\rho k^\alpha/\alpha}\,dk$.

Lemma 4. *(i) For each realized output profile,*

$$\sum_{i=1}^{n} r_i(\mathbf{y}) = -\frac{1}{2}\sum_{i=1}^{n}(y_i - \bar{y})^2 + \frac{1}{2}\sum_{i=1}^{n}(y_i - \bar{y} - \gamma)_+^2 \leq 0. \tag{11}$$

(ii) Displaying parameter dependence explicitly, let $W(x,\theta,\rho,\gamma)$ and $\pi(x,\theta,\rho,\gamma)$ denote total expected employee welfare and expected firm profit when every employee chooses x, and let $\widetilde{P}(x;\theta,\rho) = (v-1)\theta\mathbb{E}[x\tau_i(x)] - \phi^f\left(1 - e^{-\rho x^\alpha/\alpha}\right) + \frac{n-1}{2n}\theta^2\mathrm{Var}(x\tau_i(x))$. Then

$$\begin{aligned} \pi(x,\theta,\rho,\gamma)/n &= \widetilde{P}(x;\theta,\rho) - \tfrac{1}{2}\mathbb{E}\left[(y_1 - \bar{y} - \gamma)_+^2\right], \\ W(x,\theta,\rho,\gamma)/n &= \theta\mathbb{E}[x\tau_i(x)] - (1/\rho + \phi^w)\left(1 - e^{-\rho x^\alpha/\alpha}\right) \\ &\quad - \frac{n-1}{2n}\theta^2\mathrm{Var}(x\tau_i(x)) + \tfrac{1}{2}\mathbb{E}\left[(y_1 - \bar{y} - \gamma)_+^2\right], \end{aligned} \tag{12}$$

where $\mathbb{E}[(y_1 - \bar{y} - \gamma)_+^2] = 0$ if and only if $\gamma \geq \frac{n-1}{n}\theta x$. If mitigation spending is $C(\rho)$ per employee, total social welfare, $\pi + W - nC(\rho)$, is $n\{v\theta\mathbb{E}[x\tau_i(x)] - (\phi^f + 1/\rho + \phi^w)\left(1 - e^{-\rho x^\alpha/\alpha}\right) - C(\rho)\}$.

Proof. (i). By (1), $r_i = \gamma(y_i - \bar{y}) - (y_i - \bar{y})^2/2 + (y_i - \bar{y} - \gamma)_+^2/2$. Summing over employees and using $\sum_i (y_i - \bar{y}) = 0$ gives (11), where the inequality holds because $0 \leq (y_i - \bar{y} - \gamma)_+ \leq |y_i - \bar{y}|$ for $\gamma \geq 0$.

(ii). At a symmetric intensity, outputs are i.i.d., so $\mathbb{E}[\sum_i (y_i - \bar{y})^2] = (n-1)\theta^2\mathrm{Var}(x\tau_i(x))$. Taking expectations in (11) and using the definitions of profit and employee welfare gives (12). Because the support of $y_1 - \bar{y}$ has supremum $\frac{n-1}{n}\theta x$, $\mathbb{E}[(y_1 - \bar{y} - \gamma)_+^2] = 0$ if and only if $\gamma \geq \frac{n-1}{n}\theta x$. Adding profit and employee welfare cancels employee payments. Subtracting mitigation spending then gives the stated social welfare. □

Proof of Lemma 2. Write $x = x^*(\theta,\rho,\gamma)$. Because the left-hand side of (4) strictly increases in x and the right-hand side weakly decreases in x, the implicit function theorem implies that $(d/d\theta)x$ has the same sign as the following partial derivative, evaluated at fixed x:

$$\begin{aligned} &(\partial/\partial\theta)\left\{\theta\left[1 + \frac{n-1}{n}\mathbb{E}\left[\left(\gamma - \frac{\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)]\right)_+\right]\right]\right\} \\ &\quad = 1 + \frac{n-1}{n}\mathbb{E}\left[\left(\gamma - \frac{2\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)]\right)\mathbf{1}\left\{\frac{\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)] < \gamma\right\}\right]. \end{aligned} \tag{13}$$

By Lemma 3(iv), mean burnout time $\mathbb{E}[\tau_i(x^*)]$ strictly decreases in $u = \rho(x^*)^\alpha/\alpha$. Since ρ is fixed, $(d/d\theta)\mathbb{E}[\tau_i(x^*)]$ has the opposite sign to $(d/d\theta)x^*$.

(i). If $\gamma = 0$, the event $\{\frac{\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)] < \gamma\}$ is empty, and the right-hand side of (13) equals one. If $\gamma > 0$, then $\gamma - \frac{2\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)] > -\gamma$ whenever $\frac{\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)] < \gamma$. Moreover, $\Pr(\frac{\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)] < \gamma) > 0$ because no coworker burns out with positive probability. Therefore, the right-hand side of (13) exceeds $1 - \frac{n-1}{n}\gamma$, which is nonnegative when $\gamma \leq \bar{\gamma}_R$. In both cases, $(d/d\theta)x^* > 0$, which establishes part (i).

(ii). Fix $\gamma \in (\bar{\gamma}_R, \theta^m - 1)$. When $\gamma > \frac{n-1}{n}\theta x$, differentiating (10) with respect to θ and using Lemma 3 gives

$$\begin{aligned}\left[(1+\rho\phi^w)(\alpha-1)x^{\alpha-2} + \left(\frac{n-1}{n}\right)^2 \theta^2 \left(1 - e^{-\rho x^\alpha/\alpha}\right)\right](d/d\theta)x \\ = 1 + \frac{n-1}{n}\gamma - 2\left(\frac{n-1}{n}\right)^2 \theta\, \mathbb{E}[x(1-\tau_i(x))],\end{aligned} \tag{14}$$

so equilibrium intensity strictly decreases in θ if and only if

$$\mathbb{E}[x(1-\tau_i(x))] > \frac{n[n+(n-1)\gamma]}{2(n-1)^2\theta}. \tag{15}$$

Because expected lost work $\mathbb{E}[x(1-\tau_i(x))]$ is zero at $x = 0$ and strictly increases to infinity in x (Lemma 3), there is a unique intensity $\bar{x}$ at which (15) holds with equality at $\theta = \theta^m$. It suffices to show that $\bar{x} < x^*(\theta^m) < n\gamma/[(n-1)\theta^m]$ for suitable ρ and ϕ^w, since then (14) applies and (15) holds at θ^m, and $(d/d\theta)x^* < 0$ and $(d/d\theta)\mathbb{E}[\tau_i(x^*)] > 0$ hold on some interval $[\bar{\theta}, \theta^m]$ by continuity and the sign relation established above.

Because $n/(n-1) < \gamma < \theta^m - 1$ and $n \geq 3$, the right-hand side of (15) at θ^m is less than $\min\{1, n\gamma/[(n-1)\theta^m]\}$. Because $\bar{x} = n[n+(n-1)\gamma]/[2(n-1)^2\theta^m] + \mathbb{E}[\bar{x}\tau_i(\bar{x})]$ and $\mathbb{E}[\bar{x}\tau_i(\bar{x})] \leq c_\alpha \rho^{-1/\alpha}$, there exists $\bar{\rho}$ such that $\bar{x} < \min\{1, n\gamma/[(n-1)\theta^m]\}$ for all $\rho > \bar{\rho}$.

Lastly, fix $\rho > \bar{\rho}$. By (4), $x^*(\theta^m)$ is continuous and strictly decreasing in ϕ^w, satisfies $x^*(\theta^m) \geq (\theta^m)^{1/(\alpha-1)} > 1 > \bar{x}$ at $\phi^w = 0$, and converges to zero as $\phi^w \to \infty$. Therefore, there exist $0 \leq \underline{\phi}^w < \overline{\phi}^w$ such that $\bar{x} < x^*(\theta^m) < n\gamma/[(n-1)\theta^m]$ for all $\phi^w \in (\underline{\phi}^w, \overline{\phi}^w)$, as desired. □

The next lemma, used for Propositions 1(iii) and 2(iii), identifies a technology level at which equilibrium work intensity switches from increasing to decreasing.

Lemma 5. *Suppose $\gamma > \bar{\gamma}_R$. There exists $\bar{\phi}^w > 0$ such that, if $\phi^w \in (0, \bar{\phi}^w)$ and $\rho > 1/\bar{\phi}^w$, there is a technology level $\bar{\theta}$, with $\bar{x} = x^*(\bar{\theta})$, for which the following statements hold: (i) $\gamma > \frac{n-1}{n}\bar{\theta}\bar{x}$, and at $\bar{\theta}$, $(d/d\theta)x^* = 0$ and $(d^2/d\theta^2)x^* < 0$; (ii) $\mathbb{E}[\tau_i(\bar{x})] > \frac{n-1}{n}\bar{\theta}\bar{x}\,\mathrm{Var}(\tau_i(\bar{x}))$; and (iii) for fixed ϕ^w, $\bar{\theta} \to \infty$ as $\rho \to \infty$.*

Proof. (i). When $\gamma > \frac{n-1}{n}\theta x^*$, (14) gives $(d/d\theta)x^* = 0$ if and only if (15) holds with equality, in which case the right-hand side of (10) equals $\theta[n+(n-1)\gamma]/(2n)$. Write $u = \rho x^\alpha/\alpha$. By Lemma 3, $u\,\mathbb{E}[1-\tau_i(x)]$ depends on x and ρ only through u and strictly increases in u, from 0 at $u = 0$ to infinity as $u \to \infty$. Let $\bar{x}$ be the unique intensity at which

$$u\,\mathbb{E}[1-\tau_i(x)] = \frac{[n+(n-1)\gamma]^2}{4\alpha(n-1)^2(\phi^w + 1/\rho)}, \tag{16}$$

and set $\bar{\theta} = n[n+(n-1)\gamma]/\{2(n-1)^2\bar{x}\,\mathbb{E}[1-\tau_i(\bar{x})]\}$. Substitution shows that $\bar{x}$ solves (10) and that (15) holds with equality at $\theta = \bar{\theta}$. Also, $\frac{n-1}{n}\bar{\theta}\bar{x} < \gamma$ if and only if $\mathbb{E}[1-\tau_i(\bar{x})] > [n+(n-1)\gamma]/[2(n-1)\gamma]$. The right-hand side is less than one because $\gamma > n/(n-1)$.

By (16), $u \geq [n+(n-1)\gamma]^2/[4\alpha(n-1)^2(\phi^w + 1/\rho)]$ at $\bar{x}$. Because $\mathbb{E}[1-\tau_i(x)] \to 1$ as $u \to \infty$ (Lemma 3), there exists $\bar{\phi}^w > 0$ such that, whenever $\phi^w + 1/\rho < 2\bar{\phi}^w$, we have $\mathbb{E}[1-\tau_i(\bar{x})] > [n+(n-1)\gamma]/[2(n-1)\gamma]$. Suppose $\phi^w < \bar{\phi}^w$ and $\rho > 1/\bar{\phi}^w$, so that $\phi^w + 1/\rho < 2\bar{\phi}^w$. Then $\gamma > \frac{n-1}{n}\bar{\theta}\bar{x}$, so $\bar{x} = x^*(\bar{\theta})$ and $(d/d\theta)x^* = 0$ at

$\bar{\theta}$. Differentiating (14) with respect to θ at $\bar{\theta}$ then gives

$$\left[(1+\rho\phi^w)(\alpha-1)\bar{x}^{\alpha-2}+\left(\frac{n-1}{n}\right)^2\bar{\theta}^2\left(1-e^{-\rho\bar{x}^\alpha/\alpha}\right)\right](d^2/d\theta^2)x^* \\ =-2\left(\frac{n-1}{n}\right)^2\mathbb{E}[\bar{x}(1-\tau_i(\bar{x}))]<0.$$

(ii). By part (i), $\mathbb{E}[1-\tau_i(\bar{x})]>1/2$, so $\frac{n-1}{n}\bar{\theta}\bar{x}\leq[n+(n-1)\gamma]/(n-1)$. Because $\mathrm{Var}(\tau_i(x))/\mathbb{E}[\tau_i(x)]\to 0$ as $u\to\infty$ (Lemma 3), the lower bound on u in part (i) implies that, after reducing $\bar{\phi}^w$ if necessary, $\mathrm{Var}(\tau_i(\bar{x}))/\mathbb{E}[\tau_i(\bar{x})]<(n-1)/[n+(n-1)\gamma]$ whenever $\phi^w+1/\rho<2\bar{\phi}^w$. Hence $\mathbb{E}[\tau_i(\bar{x})]>\frac{n-1}{n}\bar{\theta}\bar{x}\,\mathrm{Var}(\tau_i(\bar{x}))$, as claimed.

(iii). Fix ϕ^w. As $\rho\to\infty$, the value of u solving (16) converges to a finite positive value, so $\bar{x}=(\alpha u/\rho)^{1/\alpha}\to 0$ and $\bar{\theta}\geq n[n+(n-1)\gamma]/[2(n-1)^2\bar{x}]\to\infty$. □

Proof of Proposition 1. (i). At $\rho=0$, no employee burns out, private cost is x^α/α, and all employees produce the same output, so $r_i=0$. Equation (4) gives $x^*=\{[1+(n-1)\gamma/n]\theta\}^{1/(\alpha-1)}$. Substituting into $W^*=n[\theta x^*-(x^*)^\alpha/\alpha]$ and differentiating gives $(d/d\theta)W^*=nx^*\{1-(n-1)\gamma/[n(\alpha-1)]\}$. This has the sign of $\bar{\gamma}_W-\gamma$.

(ii). Fix $\gamma>0$ and $\phi^w>0$, and write $x=x^*(\theta)$. When $\gamma>\frac{n-1}{n}\theta x$, Lemma 4 gives $W^*/n=\theta\mathbb{E}[x\tau_i(x)]-(1/\rho+\phi^w)\left(1-e^{-\rho x^\alpha/\alpha}\right)-\frac{n-1}{2n}\theta^2\mathrm{Var}(x\tau_i(x))$. Differentiating W^* with respect to θ, using the marginal expected cost $(1+\rho\phi^w)x^{\alpha-1}e^{-\rho x^\alpha/\alpha}$, (10), and Lemma 3, gives

$$\frac{1}{n}(d/d\theta)W^*=\mathbb{E}[x\tau_i(x)]-\frac{n-1}{n}\theta\,\mathrm{Var}(x\tau_i(x))-\frac{n-1}{n}\theta e^{-\rho x^\alpha/\alpha}\left[\gamma+\frac{\theta}{n}\mathbb{E}[x(1-\tau_i(x))]\right](d/d\theta)x, \tag{17}$$

where the first two terms are the direct technology effect and the last term is the strategic work-intensity effect.

Set $\theta=1$. By (4), $(x^*)^{\alpha-1}\leq[1+(n-1)\gamma/n]/(1+\rho\phi^w)$, so $x^*\to 0$ and $\rho(x^*)^\alpha\to 0$ as $\rho\to\infty$. Lemma 3 then gives $\mathbb{E}[x^*\tau_i(x^*)]/x^*\to 1$, $\mathrm{Var}(x^*\tau_i(x^*))/x^*\to 0$, and $1-e^{-\rho(x^*)^\alpha/\alpha}\to 0$. Also, $\gamma>\frac{n-1}{n}x^*$ for large ρ. Using $\mathbb{E}[x^*(1-\tau_i(x^*))]\leq x^*$ in (10) and (14) gives $(1+\rho\phi^w)(x^*)^{\alpha-1}\to 1+(n-1)\gamma/n$ and $(1/x^*)(d/d\theta)x^*\to 1/(\alpha-1)$.

Now suppose $\gamma>\bar{\gamma}_W$. Substituting these limits into (17) gives $(1/(nx^*))(d/d\theta)W^*\to 1-(n-1)\gamma/[n(\alpha-1)]<0$ at $\theta=1$. The direct technology effect divided by x^* tends to one, while $(d/d\theta)x^*>0$ for large ρ. By continuity, these signs hold on $[1,\theta_1]$ for some $\theta_1\in(1,\theta^m)$, proving part (ii).

(iii). Fix $\gamma>\bar{\gamma}_W$, which exceeds $\bar{\gamma}_R$ because $\alpha>2$, let $\bar{\phi}^w$ be the threshold in Lemma 5, and fix $\phi^w\in(0,\bar{\phi}^w)$. For $\rho>1/\bar{\phi}^w$, let $\bar{\theta}$ and $\bar{x}=x^*(\bar{\theta})$ be as in that lemma. At $\bar{\theta}$, $(d/d\theta)x^*=0$, so (17) gives $(1/n)(d/d\theta)W^*=\bar{x}\{\mathbb{E}[\tau_i(\bar{x})]-\frac{n-1}{n}\bar{\theta}\bar{x}\,\mathrm{Var}(\tau_i(\bar{x}))\}>0$. Because $(d^2/d\theta^2)x^*<0$ at $\bar{\theta}$, by continuity there exist θ_2 and θ^m with $\bar{\theta}<\theta_2<\theta^m$ such that, on $[\theta_2,\theta^m]$, $\gamma>\frac{n-1}{n}\theta x^*$, $(d/d\theta)x^*<0$, and $\mathbb{E}[x^*\tau_i(x^*)]-\frac{n-1}{n}\theta\,\mathrm{Var}(x^*\tau_i(x^*))>0$. Equation (17) therefore implies $(d/d\theta)W^*>0$ on this interval, while $(d/d\theta)x^*<0$ implies $(d/d\theta)\mathbb{E}[\tau_i(x^*)]>0$.

Lastly, for sufficiently large ρ, Lemma 5 gives $\bar{\theta}>1$, and part (ii) gives $(d/d\theta)W^*<0$ on $[1,\theta_1]$ for some $\theta_1\in(1,\bar{\theta})$. Continuity gives a minimum of W^* on $[1,\theta^m]$. Since $(d/d\theta)W^*<0$ at $\theta=1$ and $(d/d\theta)W^*>0$ at $\theta=\theta^m$, every minimum is interior. □

Proof of Proposition 2. (i). In the absence of burnout, $r_i=0$ and $\Pi^*=n(v-1)\theta x^*$. Using x^* from the proof of Proposition 1(i) gives $(d/d\theta)\Pi^*=n\alpha(v-1)x^*/(\alpha-1)>0$. This strictly increases in θ because x^* does.

(ii). Fix $\gamma > 0$ and $\phi^w > 0$, write $x = x^*(\theta)$, and consider $\theta = 1$ as $\rho \to \infty$. The proof of Proposition 1(ii) gives $x \to 0$, $\rho x^\alpha \to 0$, $(1+\rho\phi^w)x^{\alpha-1} \to 1 + (n-1)\gamma/n$, and $(1/x)(d/d\theta)x \to 1/(\alpha-1)$. Differentiating (14) with respect to θ then yields $(1/x)(d^2/d\theta^2)x \to (2-\alpha)/(\alpha-1)^2$ at $\theta = 1$, as in the absence of burnout.

Next, expected output per employee is $\mathbb{E}[y_i] = \theta\mathbb{E}[x\tau_i(x)]$. By Lemma 3, $(d/d\theta)\mathbb{E}[y_i] = \mathbb{E}[x\tau_i(x)] + \theta e^{-\rho x^\alpha/\alpha}(d/d\theta)x$ and $(d^2/d\theta^2)\mathbb{E}[y_i] = e^{-\rho x^\alpha/\alpha}[2(d/d\theta)x + \theta(d^2/d\theta^2)x - \rho\theta x^{\alpha-1}\big((d/d\theta)x\big)^2]$, so at $\theta = 1$, $(1/x)(d/d\theta)\mathbb{E}[y_i] \to \alpha/(\alpha-1) > 0$ and $(1/x)(d^2/d\theta^2)\mathbb{E}[y_i] \to \alpha/(\alpha-1)^2 > 0$. Fix sufficiently large ρ. By continuity, $(d/d\theta)\mathbb{E}[y_i] > 0$, $(d^2/d\theta^2)\mathbb{E}[y_i] > 0$, and $\gamma > \frac{n-1}{n}\theta x$ on $[1,\theta_1]$ for some $\theta_1 \in (1,\theta^m)$. On this interval, Lemma 4 gives $\Pi^*/n = (v-1)\mathbb{E}[y_i] - \phi^f\left(1 - e^{-\rho x^\alpha/\alpha}\right) + \frac{n-1}{2n}\theta^2\mathrm{Var}(x\tau_i(x))$. Because $x = x^*(\theta)$ does not depend on v, the burnout-cost term $\phi^f\left(1 - e^{-\rho x^\alpha/\alpha}\right)$ and variance term $\frac{n-1}{2n}\theta^2\mathrm{Var}(x\tau_i(x))$ have bounded first and second derivatives with respect to θ, independent of v. Both $(d/d\theta)\mathbb{E}[y_i]$ and $(d^2/d\theta^2)\mathbb{E}[y_i]$ have positive minima on $[1,\theta_1]$. Therefore, $(d/d\theta)\Pi^* > 0$ and $(d^2/d\theta^2)\Pi^* > 0$ on $[1,\theta_1]$ for sufficiently large v, proving part (ii).

(iii). Fix $\gamma > \bar{\gamma}_R$, let $\bar{\phi}^w$ be the threshold in Lemma 5, and fix $\phi^w \in (0,\bar{\phi}^w)$. For $\rho > 1/\bar{\phi}^w$, let $\bar{\theta}$ be as in that lemma. Because $(d/d\theta)x^* = 0$ and $(d^2/d\theta^2)x^* < 0$ at $\bar{\theta}$, the formulas in part (ii) reduce to $(d/d\theta)\mathbb{E}[y_i] = \mathbb{E}[x^*\tau_i(x^*)] > 0$ and $(d^2/d\theta^2)\mathbb{E}[y_i] = \bar{\theta}e^{-\rho(x^*)^\alpha/\alpha}(d^2/d\theta^2)x^* < 0$ at $\bar{\theta}$. Choose ρ large enough that $\bar{\theta} > 1$ and the argument in part (ii) applies on $[1,\theta_1]$ for some $\theta_1 \in (1,\bar{\theta})$. By continuity, there exist θ_2 and θ^m with $\bar{\theta} < \theta_2 < \theta^m$ such that, on $[\theta_2,\theta^m]$, $\gamma > \frac{n-1}{n}\theta x^*$, $(d/d\theta)\mathbb{E}[y_i] > 0$, and $(d^2/d\theta^2)\mathbb{E}[y_i] < 0$. As in part (ii), choosing v sufficiently large gives $(d/d\theta)\Pi^* > 0$ on both intervals, $(d^2/d\theta^2)\Pi^* > 0$ on $[1,\theta_1]$, and $(d^2/d\theta^2)\Pi^* < 0$ on $[\theta_2,\theta^m]$. Lastly, continuity gives a maximum of $(d/d\theta)\Pi^*$ on $[1,\theta^m]$. Every maximum is interior because $(d/d\theta)\Pi^*$ increases near 1 and decreases near θ^m. □

The next two lemmas characterize implementable intensities and the firm's optimal intensity.

Lemma 6. *(i) Equation (18) defines a unique $\gamma(x,\theta,\rho) \geq 0$ if and only if $x \geq [\theta/(1+\rho\phi^w)]^{1/(\alpha-1)}$:*

$$\mathbb{E}\left[\left(\gamma - \frac{\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)]\right)_+\right] = \frac{n[(1+\rho\phi^w)x^{\alpha-1} - \theta]}{(n-1)\theta}. \tag{18}$$

If $x = [\theta/(1+\rho\phi^w)]^{1/(\alpha-1)}$, then $\gamma(x,\theta,\rho) = 0$. If $x > [\theta/(1+\rho\phi^w)]^{1/(\alpha-1)}$, then $\gamma(x,\theta,\rho) > 0$ and $(\partial/\partial x)\gamma(x,\theta,\rho) > 0$.

(ii) If

$$(1+\rho\phi^w)x^{\alpha-1} > \theta + \left(\frac{n-1}{n}\right)^2\theta^2\mathbb{E}[x\tau_i(x)], \tag{19}$$

then $\gamma(x,\theta,\rho) > \frac{n-1}{n}\theta x$.

Proof. (i). Rearranging (4) gives (18). The left-hand side of (18) equals zero at $\gamma = 0$ and converges to infinity as $\gamma \to \infty$. For $\gamma > 0$,

$$(\partial/\partial\gamma)\mathbb{E}\left[\left(\gamma - \frac{\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)]\right)_+\right] = \Pr\left(\frac{\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)] < \gamma\right) > 0,$$

because no coworker burns out with positive probability. Therefore, (18) has a unique solution $\gamma \geq 0$ if and

only if its right-hand side is nonnegative, and this solution is zero if and only if the right-hand side is zero. For $x > [\theta/(1+\rho\phi^w)]^{1/(\alpha-1)}$, the right-hand side strictly increases in x, whereas the left-hand side weakly decreases in x (Lemma 3), so implicitly differentiating (18) gives $(\partial/\partial x)\gamma(x,\theta,\rho) > 0$.

(ii). Solving (10) for γ and using $\mathbb{E}[x(1-\tau_i(x))] = x - \mathbb{E}[x\tau_i(x)]$ shows that the solution exceeds $\frac{n-1}{n}\theta x$ exactly when (19) holds. It then also solves (18), so uniqueness proves the claim. □

We say that an intensity x is *implementable* if $x \geq [\theta/(1+\rho\phi^w)]^{1/(\alpha-1)}$, in which case the incentive $\gamma(x,\theta,\rho)$ induces x as the equilibrium intensity (Lemma 6). By Lemma 3, $(\partial/\partial x)\widetilde{P}(x;\theta,\rho) = e^{-\rho x^\alpha/\alpha}G(x;\theta,\rho)$, where

$$G(x;\theta,\rho) = (v-1)\theta + \frac{n-1}{n}\theta^2\mathbb{E}[x(1-\tau_i(x))] - \phi^f\rho x^{\alpha-1}. \tag{20}$$

By Lemma 4, profit per employee at an implementable intensity x is at most $\widetilde{P}(x;\theta,\rho)$, with equality when $\gamma(x,\theta,\rho) \geq \frac{n-1}{n}\theta x$. Consequently, if $\widetilde{P}(\cdot\,;\theta,\rho)$ has a unique maximizer $\hat{x}$ that satisfies (19), then $\hat{x}$ is implementable, $\gamma(\hat{x},\theta,\rho) > \frac{n-1}{n}\theta\hat{x}$ (Lemma 6), and $\hat{x}$ uniquely maximizes profit over implementable intensities.

Lemma 7. *(i) There exists $\kappa > 0$, depending only on α, v, and θ^m, such that if $\phi^f \max\{\rho, \rho^{1/\alpha}\} < \kappa$, then for every $\theta \in [1,\theta^m]$, $G(\cdot\,;\theta,\rho)$ has a unique root x^*, at which $(\partial/\partial x)G < 0$. This root uniquely maximizes profit over implementable intensities, and the corresponding incentive satisfies $\gamma^* > \frac{n-1}{n}\theta x^*$.*

(ii) For fixed θ and ϕ^f, as $\rho \to 0$, $\rho^{1/(\alpha-2)}x^* \to [(n-1)\theta^2/(n\phi^f)]^{1/(\alpha-2)}$ and $(\rho/x^*)(d/d\rho)x^* \to -1/(\alpha-2)$.

Proof. (i). Fix $\theta \in [1,\theta^m]$. Since $G(0;\theta,\rho) = (v-1)\theta > 0$ and $G(x;\theta,\rho) \to -\infty$ as $x \to \infty$, G has a root. At any root, $\phi^f\rho x^{\alpha-1} \geq (v-1)\theta \geq v-1$, so by Lemma 3,

$$x \geq \left[\frac{v-1}{\phi^f\rho}\right]^{1/(\alpha-1)}, \qquad \frac{\mathbb{E}[x\tau_i(x)]}{x} \leq c_\alpha\left[\frac{\phi^f\rho^{1/\alpha}}{v-1}\right]^{1/(\alpha-1)}. \tag{21}$$

The bounds on x and $\mathbb{E}[x\tau_i(x)]/x$ do not depend on θ. As $\phi^f\max\{\rho,\rho^{1/\alpha}\} \to 0$, $x \to \infty$ and $\mathbb{E}[x\tau_i(x)]/x \to 0$ at every root, uniformly over $\theta \in [1,\theta^m]$. At any root, $G = 0$ also gives $\phi^f\rho x^{\alpha-2} \geq \frac{n-1}{n}\theta^2\{1-\mathbb{E}[x\tau_i(x)]/x\}$, so, because $\left(1-e^{-\rho x^\alpha/\alpha}\right) \leq 1$,

$$(\partial/\partial x)G = \frac{n-1}{n}\theta^2\left(1-e^{-\rho x^\alpha/\alpha}\right) - (\alpha-1)\phi^f\rho x^{\alpha-2} \leq \frac{n-1}{n}\theta^2\left[1-(\alpha-1)\left(1-\frac{\mathbb{E}[x\tau_i(x)]}{x}\right)\right],$$

which is negative whenever $\mathbb{E}[x\tau_i(x)]/x < (\alpha-2)/(\alpha-1)$. By (21), this holds at every root when $\phi^f\max\{\rho,\rho^{1/\alpha}\}$ is small, so G changes sign only from positive to negative and has a unique root x^*, which uniquely maximizes $\widetilde{P}$ because $(\partial/\partial x)\widetilde{P} = e^{-\rho x^\alpha/\alpha}G$.

At $x = x^*$, divide both sides of (19) by x. The left-hand side is at least $x^{\alpha-2}$, and the right-hand side is at most $\theta^m/x + (\theta^m)^2\mathbb{E}[x\tau_i(x)]/x$. By (21), the left-hand side tends to infinity and the right-hand side to zero as $\phi^f\max\{\rho,\rho^{1/\alpha}\} \to 0$, uniformly over $\theta \in [1,\theta^m]$. The paragraph preceding the lemma then gives implementability, $\gamma^* > \frac{n-1}{n}\theta x^*$, and a unique profit maximum at x^*.

(ii). Fix θ and ϕ^f, and let $\rho \to 0$. By (21), $x^* \to \infty$ and $\mathbb{E}[x^*\tau_i(x^*)]/x^* \to 0$, and $\rho(x^*)^\alpha \geq [(v-1)/\phi^f]^{\alpha/(\alpha-1)}\rho^{-1/(\alpha-1)} \to \infty$, so $1-e^{-\rho(x^*)^\alpha/\alpha} \to 1$ at $x = x^*$. Dividing $G(x^*;\theta,\rho) = 0$ by x^* gives $\phi^f\rho(x^*)^{\alpha-2} \to \frac{n-1}{n}\theta^2$, which proves the first limit. It also gives $(\partial/\partial x)G \to -(\alpha-2)\frac{n-1}{n}\theta^2$. By Lemma 3, $\rho(\partial/\partial\rho)\mathbb{E}[x\tau_i(x)]/x \to 0$ at $x = x^*$. Using these limits when implicitly differentiating $G(x^*;\theta,\rho) = 0$ gives

$(\rho/x^*)(d/d\rho)x^* \to -1/(\alpha-2)$. □

Proof of Proposition 3. (i). Fix $\gamma > 0$ and write $x = x^*(\theta, \rho, \gamma)$. At a fixed $x > 0$, increasing ρ strictly raises the left-hand side of (4) and weakly lowers the right-hand side, because each coworker's lost work $x[1-\tau_j(x)]$ is stochastically increasing in ρ (Lemma 3). Therefore, equilibrium intensity strictly decreases in ρ. As $\rho \to 0$, (4) gives $x \to \{[1+(n-1)\gamma/n]\theta\}^{1/(\alpha-1)}$, and conditioning on which coworkers burn out shows that $\mathbb{E}[(\gamma - \frac{\theta}{n}\sum_{j=1}^{n-1} x[1-\tau_j(x)])_+]$ is continuously differentiable in (x,ρ) at $\rho = 0$, so by the implicit function theorem, $(d/d\rho)x$ remains bounded near $\rho = 0$. Therefore, $(d/d\rho)(\rho x^\alpha/\alpha) = x^\alpha/\alpha + \rho x^{\alpha-1}(d/d\rho)x \to \alpha^{-1}\{[1+(n-1)\gamma/n]\theta\}^{\alpha/(\alpha-1)} > 0$. By Lemma 3(iv), mean burnout time $\mathbb{E}[\tau_i(x)]$ strictly decreases in ρ near $\rho = 0$, proving part (i).

(ii). *Step 1.* For small ρ, Lemma 7 gives a unique optimal intensity $x^* = x^*(\theta,\rho)$ with $\gamma^* > \frac{n-1}{n}\theta x^*$, so Lemma 4 gives $W^*/n = \theta\mathbb{E}[x^*\tau_i(x^*)] - (1/\rho + \phi^w)\left(1 - e^{-\rho(x^*)^\alpha/\alpha}\right) - \frac{n-1}{2n}\theta^2\mathrm{Var}(x^*\tau_i(x^*))$. Let $u^* = \rho(x^*)^\alpha/\alpha$ denote cumulative exposure. By Lemma 7, $\rho^{1/(\alpha-2)}x^* \to [(n-1)\theta^2/(n\phi^f)]^{1/(\alpha-2)}$ and $(\rho/x^*)(d/d\rho)x^* \to -1/(\alpha-2)$. Since $u^* = \rho(x^*)^\alpha/\alpha$, we have $\rho^{2/(\alpha-2)}u^* \to \alpha^{-1}[(n-1)\theta^2/(n\phi^f)]^{\alpha/(\alpha-2)} > 0$ as $\rho \to 0$ and

$$\frac{\rho}{u^*}(d/d\rho)u^* = 1 + \alpha\frac{\rho}{x^*}(d/d\rho)x^* \longrightarrow -\frac{2}{\alpha-2}. \tag{22}$$

Because $\rho^{1/(\alpha-2)}x^*$ and $\rho^{2/(\alpha-2)}u^*$ both converge to positive constants, the product of e^{-u^*} with any fixed powers of x^* and ρ, or with the bounded elasticity $(\rho/x^*)(d/d\rho)x^*$, converges to zero as $\rho \to 0$.

Step 2. Differentiating W^* with respect to ρ along $x^*(\theta,\rho)$ and multiplying by ρ^2/n, we have $(\rho^2/n)(d/d\rho)W^* = 1 - e^{-u^*} + R(\rho)$, where

$$R(\rho) = \theta\rho^2(d/d\rho)\mathbb{E}[x^*\tau_i(x^*)] - (1+\rho\phi^w)\rho e^{-u^*}(d/d\rho)u^* - \frac{n-1}{2n}\theta^2\rho^2(d/d\rho)\mathrm{Var}(x^*\tau_i(x^*)).$$

Expand the derivatives in $R(\rho)$ using the identities in the proof of Lemma 3. Terms containing e^{-u^*} converge to zero by Step 1. Using $\mathrm{Var}(x^*\tau_i(x^*)) \le \mathbb{E}[(x^*\tau_i(x^*))^2]$, the contribution without e^{-u^*}, $(\rho/\alpha)\{-\theta\mathbb{E}[x^*\tau_i(x^*)] + \frac{n-1}{n}\theta^2\mathrm{Var}(x^*\tau_i(x^*))\}$, is bounded in absolute value by a constant multiple of $\rho\mathbb{E}[x^*\tau_i(x^*)] + \rho\mathbb{E}[(x^*\tau_i(x^*))^2] \le 2c_\alpha\rho^{1-2/\alpha}$ for $\rho \le 1$, which converges to zero because $\alpha > 2$. Therefore, $(\rho^2/n)(d/d\rho)W^* \to 1$, so W^* strictly increases in ρ near $\rho = 0$.

Step 3. Lemma 7 and (22) give $(d/d\rho)x^* < 0$ and $(d/d\rho)u^* < 0$ near $\rho = 0$, so x^* strictly increases and mean burnout time $\mathbb{E}[\tau_i(x^*)]$ strictly decreases as ρ decreases, by Lemma 3(iv).

Step 4. Solving (10) for γ gives $\gamma^* = \frac{n-1}{n}\theta\,\mathbb{E}[x^*(1-\tau_i(x^*))] + \frac{n(1+\rho\phi^w)(x^*)^{\alpha-1}}{(n-1)\theta} - \frac{n}{n-1}$. Since $x^* \to \infty$ and $\alpha > 2$, this gives $\gamma^*/(x^*)^{\alpha-1} \to n/[(n-1)\theta]$. Differentiating the formula for γ^* and using Lemmas 3 and 7 gives $(\rho/\gamma^*)(d/d\rho)\gamma^* \to -(\alpha-1)/(\alpha-2)$. The contribution from expected lost work tends to zero in this limit: $|\rho(\partial/\partial\rho)\mathbb{E}[x\tau_i(x)]| \le x/\alpha$, $(\rho/x^*)(d/d\rho)x^*$ is bounded, and $x^*/\gamma^* \to 0$. Thus, γ^* strictly decreases in ρ near zero. All four derivative signs therefore hold for sufficiently small $\rho > 0$, proving part (ii).

(iii). Suppose $\frac{(v-1)\phi^w}{\alpha} < \phi^f < (v-1)\phi^w$. Because $\left(1 - e^{-\rho x^\alpha/\alpha}\right) \le \min\{\rho x^\alpha/\alpha, 1\} \le (\rho x^\alpha/\alpha)^{(\alpha-2)/\alpha}$, (20) gives $(\partial/\partial x)G \le x^{\alpha-2}[\frac{n-1}{n}\theta^2(\rho/\alpha)^{(\alpha-2)/\alpha} - (\alpha-1)\phi^f\rho] < 0$ for sufficiently large ρ. Because $G(0;\theta,\rho) > 0$ and $G(x;\theta,\rho) \to -\infty$ as $x \to \infty$, G has a unique root x, which uniquely maximizes $\widetilde{P}$. At this root, $(v-1)\theta \le \phi^f\rho x^{\alpha-1} \le (v-1)\theta + \frac{n-1}{n}\theta^2 x$, so $x \to 0$ and $\phi^f\rho x^{\alpha-1} \to (v-1)\theta$ as $\rho \to \infty$. Consequently,

$(1+\rho\phi^w)x^{\alpha-1}-\theta-(\frac{n-1}{n})^2\theta^2\mathbb{E}[x\tau_i(x)]$ converges to $\theta[(v-1)\phi^w-\phi^f]/\phi^f>0$, so by the paragraph preceding Lemma 7, for sufficiently large ρ, $x=x^*(\theta,\rho)$ is the firm's unique optimum and $\gamma^*>\frac{n-1}{n}\theta x$.

Next, $u=\rho x^\alpha/\alpha=x\cdot\rho x^{\alpha-1}/\alpha\to 0$. Because $0\le\rho(\partial/\partial\rho)\mathbb{E}[x(1-\tau_i(x))]\le x/\alpha$ (Lemma 3), at the optimum we have $\rho(\partial/\partial\rho)G\to-(v-1)\theta$ and $x(\partial/\partial x)G\to-(\alpha-1)(v-1)\theta$. Implicit differentiation of $G(x;\theta,\rho)=0$ therefore gives $(\rho/x)(d/d\rho)x\to-1/(\alpha-1)$ and $(\rho/u)(d/d\rho)u=1+\alpha(\rho/x)(d/d\rho)x\to-1/(\alpha-1)$, so work intensity strictly decreases and mean burnout time $\mathbb{E}[\tau_i(x)]$ strictly increases in ρ for sufficiently large ρ, by Lemma 3(iv). Lastly, by Lemma 4, $W^*/n=\theta\mathbb{E}[x\tau_i(x)]-\phi^w(1-e^{-u})-R$ with $R=(1-e^{-u})/\rho+\frac{n-1}{2n}\theta^2\mathrm{Var}(x\tau_i(x))$. Because $u\to 0$, Lemma 3, $(\rho/x)(d/d\rho)x\to-1/(\alpha-1)$, and $\phi^f\rho x^{\alpha-1}\to(v-1)\theta$ imply that $(\rho/\mathbb{E}[x\tau_i(x)])(d/d\rho)\mathbb{E}[x\tau_i(x)]\to-1/(\alpha-1)$ and $\{\rho e^{-u}/(1-e^{-u})\}(d/d\rho)u\to-1/(\alpha-1)$, and that $\phi^w(1-e^{-u})/(\theta x)\to(v-1)\phi^w/(\alpha\phi^f)$, while $(1-e^{-u})/(\rho x)\le x^{\alpha-1}/\alpha\to 0$ and $0\le(\partial/\partial x)\mathrm{Var}(x\tau_i(x))\le 2x$ and $|\rho(\partial/\partial\rho)\mathrm{Var}(x\tau_i(x))|\le 4x^2/\alpha$ by Lemma 3. Together with $(\rho/x)(d/d\rho)x\to-1/(\alpha-1)$, the bounds on $(1-e^{-u})/(\rho x)$, $(\partial/\partial x)\mathrm{Var}(x\tau_i(x))$, and $\rho(\partial/\partial\rho)\mathrm{Var}(x\tau_i(x))$ imply $(\rho/x)(d/d\rho)R\to 0$. Therefore,

$$\frac{\rho}{n\theta x}(d/d\rho)W^*\longrightarrow-\frac{\alpha\phi^f-(v-1)\phi^w}{\alpha(\alpha-1)\phi^f}<0.$$

For sufficiently large ρ, $(d/d\rho)x^*<0$, $(d/d\rho)\mathbb{E}[\tau_i(x^*)]>0$, and $(d/d\rho)W^*<0$, proving part (iii).

□

Proof of Corollary 1. By Lemma 7, for sufficiently small $\rho>0$ and all $\theta\in[1,\theta^m]$, the firm's optimal intensity x^* is the unique root of $G(\cdot\,;\theta,\rho)$ and $(\partial/\partial x)G<0$ at x^*. Because $(\partial/\partial\theta)G=(v-1)+2\frac{n-1}{n}\theta\mathbb{E}[x^*(1-\tau_i(x^*))]>0$, the implicit function theorem gives $(d/d\theta)x^*>0$. Because mean burnout time strictly decreases in $u=\rho(x^*)^\alpha/\alpha$ (Lemma 3(iv)) and ρ is fixed, $\mathbb{E}[\tau_i(x^*)]$ strictly decreases in θ, as desired. □

For fixed exposure ρ, let $x^S(\rho)$ and $\gamma^S(\rho)$ denote the social design's optimal work intensity and incentive, and let $\tau_i^S(\rho)$ denote employee i's associated censored burnout time.

Lemma 8. *(i) At any fixed exposure $\rho>0$, the social design chooses*

$$x^S(\rho)=\max\left\{\left[\frac{\theta}{1+\rho\phi^w}\right]^{1/(\alpha-1)},\left[\frac{v\theta}{1+\rho(\phi^w+\phi^f)}\right]^{1/(\alpha-1)}\right\}. \tag{23}$$

(ii) Its incentive is zero if $\rho[\phi^f-(v-1)\phi^w]\ge v-1$, and strictly positive otherwise.

Proof. (i). By Lemma 6, implementable intensities are those at least $[\theta/(1+\rho\phi^w)]^{1/(\alpha-1)}$. Using Lemmas 3 and 4, the derivative of social welfare per employee with respect to x is

$$e^{-\rho x^\alpha/\alpha}\left[v\theta-\{1+\rho(\phi^w+\phi^f)\}x^{\alpha-1}\right].$$

This changes sign once, from positive to negative. Taking the larger of its root and the lower bound on x gives (23).

(ii). Comparing the two terms in (23) gives the stated condition. By Lemma 6, the incentive is zero at the lower bound and positive above it. □

Proof of Proposition 4. Set $\underline{\theta}=\min\{2,\theta^m\}$ and fix $\theta\in[1,\underline{\theta}]$.

Step 1. Take ϕ^w sufficiently large and $\phi^f > 0$ sufficiently small that $(v-1)\phi^w \geq \phi^f$. By Lemma 8, the intensity that maximizes social welfare at each exposure is then $x^S(\rho) = [v\theta/\{1+\rho(\phi^w+\phi^f)\}]^{1/(\alpha-1)}$. Because $1+\rho(\phi^w+\phi^f) \geq \max\{1, \rho\phi^w\}$, mean burnout time under the social design satisfies $1-\mathbb{E}[\tau_i^S(\rho)] \leq \rho x^S(\rho)^\alpha/\alpha \leq (v\theta)^{\alpha/(\alpha-1)}/(\alpha\phi^w)$, using $1-e^{-z} \leq z$ in the expectation formula in Lemma 3(iv). Because $x[1-\tau_j(x)] \leq x$ for each coworker j, the left-hand side of (18) is at least $\gamma - \frac{n-1}{n}\theta x$, so (18) and (23) give $\gamma^S \leq \frac{n-1}{n}\theta x^S + n[(1+\rho\phi^w)(x^S)^{\alpha-1} - \theta]/[(n-1)\theta] \leq \frac{n-1}{n}\theta(v\theta)^{1/(\alpha-1)} + n(v-1)/(n-1)$. Lastly, at $\gamma = \gamma^S(\rho)$, the social value per employee before mitigation spending, $[\Pi^* + W^*]/n$, is at most $v\theta\mathbb{E}[x^S\tau_i(x^S)] \leq v\theta x^S(\rho)$.

Step 2. For every incentive γ and induced intensity x, Lemmas 3 and 4 give

$$\Pi^*/n \leq \widetilde{P}(x;\theta,\rho) \leq (v-1)\theta\mathbb{E}[x\tau_i(x)] + \frac{n-1}{2n}\theta^2\mathbb{E}[(x\tau_i(x))^2] \leq c_\alpha\left[(v-1)\theta\rho^{-1/\alpha} + \theta^2\rho^{-2/\alpha}\right], \tag{24}$$

which does not depend on ϕ^w or ϕ^f and vanishes as $\rho \to \infty$. By Lemma 3 and Assumption 1, $\rho^{1/\alpha}\mathbb{E}[\tau_i(1)]$ converges to a positive constant and $\rho C(\rho) \to 0$ as $\rho \to \infty$, so there exists ρ_0 such that $C(\rho_0) < (v-1)\mathbb{E}[\tau_i(1)]/2$, where the expectation here and below is evaluated at exposure ρ_0. If $1+\rho_0\phi^w \geq \underline{\theta} + \underline{\theta}^2$, then (19) holds at $(x,\rho) = (1,\rho_0)$, so by Lemmas 6 and 4, and because $\theta \geq 1$, $\left(1-e^{-\rho x^\alpha/\alpha}\right) \leq 1$, and $\mathrm{Var}(x\tau_i(x)) \geq 0$, implementing $x=1$ at exposure ρ_0 yields $\Pi^*/n - C(\rho_0) \geq (v-1)\mathbb{E}[\tau_i(1)] - \phi^f - C(\rho_0) > (v-1)\mathbb{E}[\tau_i(1)]/4 > 0$ for sufficiently small ϕ^f. Lastly, choose $\rho_1 \geq \rho_0$ such that the right-hand side of (24) at $\theta = \underline{\theta}$ is below this positive lower bound for all $\rho > \rho_1$. Note that ρ_1 depends only on $\underline{\theta}$, v, α, and C.

Step 3. For $\rho \in (0, \rho_1]$, we have $\phi^f \max\{\rho, \rho^{1/\alpha}\} \leq \phi^f \max\{\rho_1, \rho_1^{1/\alpha}\}$, so for sufficiently small ϕ^f, Lemma 7 implies that the firm's optimal intensity at exposure ρ is the unique root $x^F(\rho)$ of $G(\cdot\,;\theta,\rho)$, implemented by an incentive greater than $\frac{n-1}{n}\theta x^F(\rho)$, and that the maximized profit per employee, $\widetilde{P}(x^F(\rho);\theta,\rho)$, is continuous in ρ because $(\partial/\partial x)G < 0$ at $x^F(\rho)$. By (24), $2/\alpha < 1$, and $\rho C(\rho) \to \infty$ as $\rho \to 0$, the maximized profit per employee net of mitigation spending, $\widetilde{P}(x^F(\rho);\theta,\rho) - C(\rho)$, converges to $-\infty$ as $\rho \to 0$. By Step 2, implementing $x=1$ at ρ_0 yields greater profit per employee net of mitigation spending than every design with $\rho > \rho_1$. Therefore, profit per employee net of mitigation spending has a maximum, and every maximizing design satisfies $\rho^F \leq \rho_1$ and $\gamma^F > \frac{n-1}{n}\theta x^F$. Lastly, $G(x^F;\theta,\rho^F) = 0$ gives $x^F \geq [(v-1)/(\phi^f\rho_1)]^{1/(\alpha-1)}$ and $\rho^F(x^F)^\alpha/\alpha \geq (v-1)x^F/(\alpha\phi^f)$, so $x^F \to \infty$ and $\rho^F(x^F)^\alpha/\alpha \to \infty$ as $\phi^f \to 0$.

Step 4. For $\rho \leq \rho_1$ and $\gamma = \gamma^S(\rho)$, Step 1 and $x^S(\rho) \leq [v\theta/(\rho\phi^w)]^{1/(\alpha-1)}$ give $\rho[(\Pi^*+W^*)/n - C(\rho)] \leq K - \rho C(\rho)$, where $K = (v\underline{\theta})^{\alpha/(\alpha-1)}(\phi^w)^{-1/(\alpha-1)}\rho_1^{(\alpha-2)/(\alpha-1)}$. Because $\rho C(\rho)$ is positive, continuous, and diverges as $\rho \to 0$, it is bounded below by a positive constant on $(0,\rho_1]$, and because ρ_1 does not depend on ϕ^w, $K - \rho C(\rho) < 0$ for all $\rho \leq \rho_1$ when ϕ^w is sufficiently large. As $\rho \to \infty$, $x^S(\rho) \to 0$ and $\rho x^S(\rho)^\alpha \to 0$, so Lemmas 3 and 4 and (23) give $[(\Pi^* + W^*)/n]/[v\theta x^S(\rho)] \to 1 - 1/\alpha$ at $\gamma = \gamma^S(\rho)$, whereas $\rho x^S(\rho) \to \infty$ because $\alpha > 2$, and $\rho C(\rho) \to 0$. Therefore, $(\Pi^* + W^*)/n - C(\rho)$ at $\gamma = \gamma^S(\rho)$ is positive for large ρ and converges to zero as $\rho \to \infty$. The social welfare objective therefore has a maximum. Every maximizing design satisfies $\rho^S > \rho_1 \geq \rho^F$ and has lower mitigation spending, $C(\rho^S) < C(\rho^F)$, because C strictly decreases.

Step 5. Step 3 and Lemma 3(iv) give $\gamma^F > \frac{n-1}{n}\theta x^F \to \infty$ and $\mathbb{E}[\tau_i^F] \to 0$ as $\phi^f \to 0$. Step 1 bounds γ^S independently of ϕ^f and gives $\mathbb{E}[\tau_i^S] \to 1$ as $\phi^w \to \infty$. These bounds hold uniformly over $\theta \in [1, \underline{\theta}]$ and all optimal designs. Together with Step 4, they allow us to choose $\underline{\phi}^w$ large enough and $\overline{\phi}^f$ small enough, with $(v-1)\underline{\phi}^w \geq \overline{\phi}^f$, that $\rho^S > \rho^F$, $\gamma^S < \gamma^F$, and $\mathbb{E}[\tau_i^S] > \mathbb{E}[\tau_i^F]$ throughout the stated parameter region. □